
MACROECONOMICS II NOTES

Econ 713*

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1 Neo-classical growth model

1.1 Planner's Problem

We start with the standard workhorse modern macroeconomic model. The Planner's problem, in recursive formulation is given by:

$$V(k) = \max_{k' \in [0, f(k) + (1-\delta)k]} \{u(f(k) + (1-\delta)k - k') + \beta V(k')\} \quad (1)$$

Where, the environment of the model is given by:

1. Large number of identical households
 - capital endowment: each household starts with k_0 unit of capital
 - capital depreciates at a rate of $\delta \in [0, 1]$
 - Labor endowment: each household chooses labour $n_t \in [0, 1]$
 - labor supplied inelastically (population size of 1)
2. Consumption good: there is a production of a single consumption good using only two inputs capital and labor.
 - The production function $F(\cdot)$ is standard neo-classical¹ $F(k, 1) = f(k)$
3. Preference of the representative agent are represented by a utility function $u(\cdot)$ ² with discounting factor $\beta \in (0, 1)$
4. State space³ is $X = [0, \max\{k_0, \bar{k}\}]$, $\bar{k}$ is maximum possible capital stock, which can be seen if household decides to consume nothing. Therefore, $\bar{k} = f(\bar{k}) + (1 - \delta)\bar{k}$

The solution of the above problem is given by a policy function g that maps from the state space to state space, that is $k' = g(k)$, and the value function $V(k)$.

There are mainly three types of numerical computational method of solving the dynamic problem, all focusing on the value function.

¹ $F(k, n)$ is continuous differentiable, strictly increasing and concave in both arguments, CRS, follows inada conditions, and inputs are essential for production.

²Utility function is strictly increasing, strictly concave, with marginal utility at 0 consumption as $+\infty$

³A state variable (vector) at any period t is smallest set of variables that determines, (1) the feasible set of the controls, (2) the current return at t given x' , and (3) the value tomorrow given

1. Value function iteration : This method proceeds by constructing a sequence of value functions and associated policy functions. The sequence is created by iterating on the recursive equation, starting from $V_0 = 0$, and continuing until V_j has converged.
2. Guess and verify : This method involves guessing and verifying a solution V to equation (1). This method relies on the uniqueness of the solution to the equation, but because it relies on luck in making a good guess, it is not generally available.
3. Howard's improvement algorithm : This method is also known as policy function iteration. In this pick a feasible policy, $g_0(x)$, and compute the value associated with operating forever with that policy. This method skips the optimization once in a while, iterate on the value function for a given g_j and then iterate over j to converge.

The solution through the first method of value function iteration can be obtained by transforming the continuous state space (of problem in (1)) to discrete state-space, and then iterating the value function on it. This can be achieved by dividing the continuous state space in intervals and replacing the state space X with $K = \{k_1, k_2, \dots, k_n\}$ with k_i as the extreme points of the successive disjoint intervals. ($k_1 < k_2 < \dots < k_n$). The transformed problem can be written as:

$$V(k) = \max_{k' \in [0, f(k) + (1-\delta)k], k' \in K} \{u(f(k) + (1-\delta)k - k') + \beta V(k')\}$$

$$\forall k \in K$$

Now the problem reduces to finding the appropriate n and K . Considering $k_0 < k^*$, where, $k^* = g(k^*) =$ we may choose $k_1 = k_0, k_n = k^*$ and setting n such that the increasing n further will keep the value in ϵ NBD.

Now, how do we know that the value function will converge for sure? Noting that the bellman equation is a contraction mapping and using the contraction mapping theorem in the Banach space, we know there will be a unique fixed point. Therefore, we can implement recursive method of successive approximation based upon:

$$V_{j+1}(k) = \max_{k' \in [0, f(k) + (1-\delta)k], k' \in K} \{u(f(k) + (1-\delta)k - k') + \beta V_j(k')\}$$

for some arbitrary initial guess V_0 .

$$\hat{V}_0 = \begin{bmatrix} V_0(k_1) \\ v_0(k_2) \\ \vdots \\ v_0(k_n) \end{bmatrix} = \begin{bmatrix} v_0^1 \\ v_0^2 \\ \vdots \\ v_0^n \end{bmatrix}$$

Similarly, the initial guess for the policy function is given by:

$$\hat{g}_j = \begin{bmatrix} g_j(k_1) \\ g_j(k_2) \\ \vdots \\ g_j(k_n) \end{bmatrix} = \begin{bmatrix} g_j^1 \\ g_j^2 \\ \vdots \\ g_j^n \end{bmatrix}$$

Now, Step- j decision rule is

$$\hat{\mathbf{g}}_j \equiv \begin{matrix} (n \times 1) \end{matrix} \begin{bmatrix} g_j(k_1) \\ g_j(k_2) \\ \vdots \\ g_j(k_n) \end{bmatrix} = \begin{bmatrix} g_j^1 \\ g_j^2 \\ \vdots \\ g_j^n \end{bmatrix}$$

Considering the full depreciation for simplicity here, when $\delta = 1$ (for all j) current pay-off function is:

$$\begin{aligned} \hat{\mathbf{F}} \equiv \begin{matrix} (n \times n) \end{matrix} & \equiv \begin{bmatrix} u(f(k_1) - k_1) & u(f(k_1) - k_2) & \cdots & u(f(k_1) - k_n) \\ u(f(k_2) - k_1) & u(f(k_2) - k_2) & \cdots & u(f(k_2) - k_n) \\ \vdots & \vdots & \ddots & \vdots \\ u(f(k_n) - k_1) & u(f(k_n) - k_2) & \cdots & u(f(k_n) - k_n) \end{bmatrix} \\ & = \begin{bmatrix} F^{11} & F^{12} & \cdots & F^{1n} \\ F^{21} & F^{22} & \cdots & F^{2n} \\ \vdots & \vdots & \ddots & \vdots \\ F^{n1} & F^{n2} & \cdots & F^{nn} \end{bmatrix} \end{aligned}$$

Now note that at any given iteration the state is determined, and therefore the transpose of the j^{th} value iteration is multiplied by the vector of $\mathbf{1}$. Hence, Transformed model can be written as

$$\hat{\mathbf{v}}_{j+1} = \max \left\{ \hat{\mathbf{F}} + \beta (\mathbf{1} \hat{\mathbf{v}}_j^\top) \right\}$$

where $\mathbf{1}_{(n \times 1)}$ is a vector of 1's, and the max operator applies line-by-line. Now we do not know how efficient this is, therefore for the efficiency consideration we will compute the F only once and using the

properties of the model such as monotone convergence of k to steady state, which implies that the policy function g is also monotone.

Howard's improvement algorithm can also be used for the efficiency consideration.

1.2 Competitive Equilibrium

In the Arrow-Debreu model, there are three kinds of agents in the economy: the households, the producers, and the market. The households are born with a positive amount of endowment and sell all their endowment to the market in the beginning and buy whatever they want to consume. The households hold proportional ownership of the producers. The profit made by the j^{th} firm is distributed among the households. Ownership is exogenous. The households possess preferences over the bundles of commodities and the preferences are standard.

Market is only capable of choosing a market price vector. Others are price takers (no bargaining behavior), the main objective of the market is to choose the prices such as the market clears for all commodities in the market. Therefore, the market is playing the role of "Walrasian auctioneer".

There are various kinds of possible decentralization, we will be discussing the Arrow-Debreu (time-0) market and sequential market. In the AD economy with time-0 markets of decentralization, market chooses sequence of time-0 price of the commodities at time t . Taking this problem in the form of recursive formulation, the state vector is now individual's capital stock (k) and aggregate capital stock (K). The price functions are $R(K)$, and $w(K)$, note that the price functions are the function of aggregate stock but individual's, implies that the individual's and firms are not price setters.

Given the above setup, the consumer's problem is the following (in recursive formulation):

$$V(k, K) = \max_{c, k'} \{u(c) + \beta V(k', K')\}$$

s.t.

$$c + k' = [R(K) + 1 - \delta]k + w(K)$$

$$c, k' \geq 0$$

with

$$K' = G(K).$$

Here $G(K)$ is policy function of the aggregate capital stock. In this case, the individual policy function is $g(k, K)$. This becomes significantly more complex in principle as there is a possibility of off-equilibrium

consumer behavior.

Definition 1. A Recursive Competitive Equilibrium (RCE) is a value function $V(k, K)$, a decision rule $g(k, K)$, an aggregate law of motion $G(K)$ and price functions $R(K), w(k)$ such that:

1. The representative firm optimizes given $R(K)$ and $w(K)$, that is

$$R(K) = F_K(k^d, n^d)$$

$$w(K) = F_N(k^d, n^d)$$

2. The representative household optimizes given $R(K)$, $w(K)$, and $G(K)$, that is $V(k, K)$ solves the household's maximization problem and $g(k, K)$ is the associated decision rule.

3. Market clears:

$$k^d = K$$

$$n^d = 1$$

4. Consistency, that is $G(k) = g(k, k)$

The last point (4) states that the consumer has rational expectations. The consumer belief about the aggregate law of motion coincides with the actual law of motion in equilibrium. The rational expectations are formulated as a rational expectations equilibrium in terms of a fixed point of an operator that maps beliefs into optimal beliefs.

The solution of the model is to find $g(k, K)$, $R(K)$, and $w(K)$ s.t.

$$u'(c(k, K)) = \beta[R(g(k, K)) + 1 - \delta]u'(c(g(k, K), g(k, K))) \quad (2)$$

$$c(k, K) + g(k, K) = w(K) + [R(K) + 1 - \delta]k \quad (3)$$

$$R(K) = F_K(K, 1) \quad (4)$$

$$w(K) = F_N(K, 1) \quad (5)$$

$$k = K \quad (6)$$

The equality (2) above is Euler equation from derived from the FOC of the household problem, since, the utility function is standard we have it to be sufficient as well. Interpretation of the equation is that while making decision or choices the household evaluates the marginal cost of saving one unit to be equal to the discounted marginal utility of consuming the return of capital.

The equation (3) is the law of motion, left hand of this equation (c and g) is a function of (k, K) and w, R are the function of K , households are not price setters.

Equation (4) and (5) are from the firm's optimization problem, firm equates marginal product of labor to the wage and marginal product of capital to the return on the capital.

We can infer from the equation (6) that in equilibrium $k = K$ since we are solving the problem of representative agent problem. Though this does not mean that the individual is price setter, in all the previous equations the individual's capital is different from the aggregate capital. This last equations helps us to restrict to look for the decision rule along the equilibrium path: which is when $G(k) = g(k, k)$, and the pricing function $R(k), w(k)$.

To see this problem in the discrete state space, we need to find G as the fixed point, the steady state. Similarly as in the value function iteration, we put an outer loop on G .

RCE with variable labor: The RCE with the variable labor would be similar with a little change in the consumer's problem and firm's optimality condition. Also, one more market is present here (labor market), therefore, the labor market also clears in the equilibrium.

Consumer's problem:

$$V(k, K) = \max_{c, k'} \{u(c, 1 - n) + \beta V(k', K')\}$$

s.t.

$$c + k' = [R(K) + 1 - \delta]k + w(K)n$$

$$c, k' \geq 0$$

$$n \in [0, 1]$$

with

$$K' = G^K(K)N = G^N(K)$$

and in this case the Firm's optimality will imply

$$R(K) = F_K(K, G^N(K))$$

$$w(K) = F_N(K, G^N(K))$$

2 Inequality and Aggregation

What we have seen in the previous model(s) that the agents are homogeneous, now consider that the agents are heterogeneous. We can introduce heterogeneity in two ways in Macroeconomics.

1. Representative agent : aggregation
In this aggregate matter for the inequality but converse is not the true. This model is more tractable and restrictive (off course).
2. No representative agent: In these kind of models there is a two way interaction, that inequality matters for aggregate and aggregate matters for inequality. Because of this two-way interaction, the problem becomes challenging, but interesting (potentially).

Consumer heterogeneity: Consumer heterogeneity can arise from various sources:

1. Wealth Endowments: Each agent in the economy differs with respect to the wealth endowment, agent born with unequal wealth. Now the inequality might or might not change over time.
2. Labor productivity: The productivity of each agent is different and constant over time. The productivity might be state dependent or may evolve over time, in both of the scenarios the agent's productivity may have dynamic nature (this can be the starting point on how do you invest in the human capital formation, or to increase in the productivity).
3. Preferences: The agent might have different preferences over leisure and consumption.

If marginal propensity of consumption is constant for all the agents then wealth distribution does not affect aggregates. Such kind of preferences are known as Gorman preferences⁴

⁴The standard consumer theory is based on one agent, but how can we use the consumer theory to say something about how our entire society is behaving. One way to think is to consider the entire society as one agent with endowment as the sum of all the agent's endowment, but under what circumstances we can represent the entire population as a single agent. Formally, $X(p, m^1, m^2, \dots, m^n) = \sum_i x_i(p, m^i) = X(p, \sum_i m^i)$. This is true under quasi linear preferences and homothetic preferences, these are also known as Gorman preferences.

2.1 *Wealth Heterogeneity: A Simple variant:*

Consider there are N individuals in the economy, where every individual i is of type $\mu^i \in [0, 1]$ such that $\sum_i \mu^i = 1$, preference of the agents are

$$U = \sum_{t=0}^{\infty} \beta^t u(c_t^i)$$

with

$$u(c) = \begin{cases} \ln(\bar{c} + c) & \bar{c} + c \geq 0 \\ (\bar{c} + c)^\pi & \bar{c} + c \geq 0 \\ -\bar{c} \exp(-\eta c) & \bar{c} + c < 0 \end{cases}$$

and the heterogeneity is with respect to wealth, which is a_0^i .

Here we will be focusing on the *log* case, note that $\bar{c}$ is subsistence consumption level with $\bar{c} \leq 0$. Therefore in this case of the utility function, the *inada* condition is same except the limit tends to $\bar{c}$.

In this model, firm owns the capital and therefore makes the investment decisions for each time period. The representative firm with $y_t = f(k_t)$. We will be considering the decentralization where firms owns the capital, consumers own firms, and Arrow-Debreu markets.

Given the structure of the decentralization, let p_t be the time zero price of y_t for all t with normalizing $p_0 = 1$. The firm solves the following problem:

$$V_0 = \max_{\{d_t, k_{t+1}\}} \sum_{t=0}^{\infty} p_t d_t$$

with k_0 given, and

$$d_t = f(k_t) - [k_{t+1} - (1 - \delta)k_t]$$

Solving for the problem will give us the FOC as,

$$p_t = p_{t+1} (f'(k_{t+1}) + (1 - \delta))$$

Interpretation: The term in the bracket of the right hand side of the above equation is the total return from investment done at time t . $f'(k_{t+1})$ can also be seen as rental rate R_{t+1} , and therefore the whole term in bracket as real interest rate $1 + r_{t+1}$. Which gives $\frac{p_t}{p_{t+1}} = (1 + r_{t+1})$, i.e., 1 unit of consumption today will cost $(1 + r_{t+1})$

tomorrow. Note that here we have this for all t , which indicates stationarity (how you value tomorrow as compared to today is same as how you value 100^{th} and 101^{th} today).

Consumer i solves the following problem:

$$\max_{\{c_t^i\}} \sum_{i=0}^{\infty} \beta^t u(c_t^i)$$

subject to the budget constraint:

$$\sum_{i=0}^{\infty} p_t c_t^i = a_0^i$$

Given the change in the wealth endowment, we have each agents has a share in the aggregate value of the firm outcome, let this share be s_0^i , therefore $a_0^i = s_0^i V_0$. Also the share of wealth is $a_0 = \sum_i \mu^i s_0^i V_0 = V_0$, and (type of the agent i) $\mu^i, s_0^i \in [0, 1]$ which comes from assuming that $\sum_i \mu^i s_0^i = 1$.

Goal: To understand the dynamics or evolution of $\{s_0^i\}_{i=1}^N$ over time. Consumer problem gives the following FOC:

$$\beta^t u'(c_t^i) = \lambda^i p_t$$

given the utility function as log, we have

$$c_t^i = \frac{\beta^t}{\lambda^i p_t} - \bar{c}$$

where, λ^i is the multiplier on the budget constraint for agent i . Substituting the one in other gives us:

$$\begin{aligned} \sum_{t=0}^{\infty} p_t \left(\frac{\beta^t}{\lambda^i p_t} - \bar{c} \right) &= a_0^i \\ \frac{1}{\lambda^i(1-\beta)} - \sum_t p_t \bar{c} &= a_0^i \\ \frac{1}{\lambda^i} &= \left(a_0^i + \sum_t p_t \bar{c} \right) (1-\beta) \\ \text{now, } c_t^i &= \frac{\beta^t(1-\beta)(a_0^i + \sum_t p_t \bar{c})}{p_t} - \bar{c} \\ c_0^i &= \frac{1}{\lambda^i} - \bar{c} = \Theta(\{p_t\}_{t=0}^{\infty}, \bar{c}) + (1-\beta)a_0^i \end{aligned}$$

The first term above (Θ) would vanish if the utility function would have been $\log(c)$, also the second term there is a constant multiplier $(1 - \beta)$ which is marginal propensity to consume.

Digression: MPS: The marginal propensity to consume is the derivative of consumption function with respect to a_0^i , here we have it as β and marginal propensity to consume is $(1 - \beta)$.

Whereas, the Average Propensity to Consume (APC) is the ratio of consumption and wealth, and average propensity to save (APS) is $1 - \text{APC}$. Therefore, in this case, we have $\text{APC} = \frac{\Theta}{a_0^i} + (1 - \beta)$. Note that though the MPC is constant, the APC is variable and dependent on the individual endowment.

Steady State: In the steady state $k' = k$. While inequality does not matter for aggregates but aggregates matters for inequality, To see this let us start with the steady state:

$$1 = \beta[f'(k) + 1 - \delta] \rightarrow f'(k) = \beta^{-1} - (1 - \delta).$$

From the Euler equation we have, the ratio of price at t to price at $(t + 1)$ is β^{-1} , that implies, $p_t = \beta^t$. This implies price declines over time and is proportional to how individuals value future periods as compared to current period. Also, the value of the overall asset is

$$a = \sum_{t=0}^{\infty} p_t d = \sum_{t=0}^{\infty} \beta^t [f(k) - \delta k] = \frac{f(k) - \delta k}{1 - \beta}$$

The aggregate consumption and asset are given as,

$$c = f(k) - \delta k$$

$$\sum_{i=1}^N \mu^i a^i = a$$

Substituting in the individual level consumption, we have

$$c^i = \bar{c}[(1 - \beta) \sum_{t=0}^{\infty} p_t - 1] + (1 - \beta)a^i = (1 - \beta)a^i$$

the above equation indicates that the consumption of the i^{th} individual is $(1 - \beta)$ proportion of the a^i in the steady state, which is $s^i V$ and also in the steady state the $MPS = APS$

All of the above equations indicates that there are multiple distributions consistent with the steady-state. However, given the initial dis-

tribution, the economy converges to a single steady-state distribution.

Now the interesting question is how the distribution evolve over time and where does it converge. For that let us obtain individual-level wealth dynamics as function of prices and saving rate from consumer's budget constraint.

$$\begin{aligned}
\sum_{\tau=t}^{\infty} p_{\tau} c_{\tau}^i &= p_t a_t^i \\
p_t c_t^i + \sum_{\tau=t+1}^{\infty} p_{\tau} c_{\tau}^i &= p_t a_t^i \\
p_t c_t^i + p_{t+1} a_{t+1}^i &= p_t a_t^i \\
\frac{c_t^i}{a_t^i} + \frac{p_{t+1} a_{t+1}^i}{p_t a_t^i} &= 1 \\
\frac{a_{t+1}^i}{a_t^i} &= \frac{p_t}{p_{t+1}} \left(1 - \frac{c_t^i}{a_t^i} \right)
\end{aligned}$$

Note that in the last equation above, left hand side is the wealth dynamics, which is equal to the product of proportion of the price ratio and APS , from before, we know that the MPC is given by

$$\frac{c_t^i}{a_t^i} = \frac{\Theta(\{p_t\}_{t=0}^{\infty}, \bar{c})}{a_t^i} + 1 - \beta$$

Knowing that $a_t^i = s_t^i a_t$, we have

$$\frac{a_{t+1}^i/a_t^i}{a_{t+1}/a_t} = \frac{1 - c_t^i/a_t^i}{1 - c_t/a_t} = \frac{s_{t+1}^i}{s_t^i}$$

The above term will be greater than 1

$$\iff \frac{c_t^i}{a_t^i} < \frac{c_t}{a_t} \iff \Theta(\{p_t\}_{t=0}^{\infty}, \bar{c})(a_t^i - a_t) > 0 \iff \Theta(\{p_t\}_{t=0}^{\infty}, \bar{c})a_t(s_t^i - 1) > 0$$

This implies the wealth dynamics of consumer i depends on the sign of Θ and sign of $(s_t - 1)$

Based on the assumptions, we have that the sequence of $f'(k_t)$ is decreasing and $\frac{p_t}{P_{t+1}}$ converges to $1/\beta$ from above. This implies that $p_t \geq \beta^t$. All these conditions together with that $\bar{c} < 0$, we have:

$$\Theta(\{p_t\}_{t=0}^{\infty}, \bar{c}) = \bar{c} \left[(1 - \beta) \sum_{t=0}^{\infty} p_t - 1 \right] \geq \bar{c} \left[(1 - \beta) \sum_{t=0}^{\infty} \beta^t - 1 \right] = 0$$

The above inequality implies that as $\bar{c} < 0$ we have $\Theta > 0$

In this case, there will be ever widening wealth distribution along transition, without mobility, the rank is preserved. This is because, $\bar{c}$ generates saving rates which is increasing in wealth, ($APC \neq MPC$). Aggregate economy is growing and induces everyone in the economy to save, though there will be decline in the interest rate. All individual including poor need to reach at the minimal consumption level $\bar{c}$, since marginal utility is very high. The one who born with higher wealth will save more and get more return, whereas the poor who born with less endowment will save very less as to reach the consumption level.

Note that the above conclusion is based on the single interest rate for all the consumers (poor or rich). To mitigate the inequality or ever widening wealth inequality with the saving interest rate different for poor and rich. This difference in the interest rate can be achieved through the credit constraints.

2.2 Competitive Equilibrium without Aggregation

In this section we will explore how the competitive equilibrium can be achieved without aggregation, relying on the Welfare theorems.

First welfare theorem Every competitive equilibrium is a Pareto optimal allocation. Whereas the *Second Welfare theorem* states that under certain assumptions Pareto optimal allocation, social welfare maximization, can be achieved as a competitive equilibrium with redistribution of the endowments.

For the benchmark model, we will solve the social planner's problem with given weights and then compute competitive equilibrium prices and ensure that those are affordable.

Consider the economy as before, endowment economy with heterogeneous agents, (two equally sized consumers). Without loss of generality, we assume that the aggregate endowment is 0 or $\sum_i a_0^i = 0$. Now the planner's problem can be formulated as follows:

$$\max_{(c_t^1, c_t^2)_{t=0}^{\infty}} \sum_t \beta^t [\alpha_1 u(c_t^1) + \alpha_2 u(c_t^2)]$$

:

Benchmark. Consider Planer solving the following problem

$$\max_{\{c_1^t, c_2^t\}_{t=-\infty}^{\infty}} \sum_{t=0}^{\infty} [\alpha_1 u^t(c_1^t) + \alpha_2 u^0(c_2^t)]$$

subject to:

$$\sum_i c_t^i = \sum_i y_t = 2y_t \forall t$$

Here α_i is the weight planner gives to agent i 's utility. The weights are not need to be add up to 1. To solve the problem, we set up the Lagrangian,

$$\mathcal{L} = \sum_{t=0}^{\infty} (\alpha_1 u_1^*(c_1^t) + \alpha_2 u_2(c_2^t)) + \sum_{t=0}^{\infty} \lambda_t \left(\sum_i y_i^t - \sum_i c_i^t \right)$$

Solving for the first order conditions, we have

$$\begin{aligned}
\alpha_1 u'_1(c_1^t) + \partial_t(-1) &= 0 \quad \forall t \\
\alpha_2 u'_2(c_2^t) + \alpha_t(-1) &= 0 \quad \forall t \\
\Rightarrow \alpha_i u'_i(c_i^t) &= \alpha_j u'_j(c_j^t) \quad \forall t \\
\Rightarrow \frac{\alpha_i}{\alpha_j} &= \frac{u'_j(c_j^t)}{u'_i(c_i^t)} \\
u'_i(c_i^t) &= \frac{\alpha_j}{\alpha_i} u'_j(c_j^t) \\
c_i^t &= (u'_i)^{-1} \left[\frac{\alpha_j}{\alpha_i} u'_j(c_j^t) \right] \\
\sum_i y_i^t &= \sum_i c_i^t = \sum_i (u'_i)^{-1} \left[\frac{\alpha_j}{\alpha_i} u'_j(c_j^t) \right] \\
\Rightarrow \sum_i (u'_i)^{-1} &\left[\frac{\alpha_j}{\alpha_i} u'_j(c_j^t) \right] = Y^t.
\end{aligned}$$

This can be seen from the above equation that c_j is a function of aggregate endowment Y^t . Consider the case when the utility is of the form logarithmic, $u = \log$.

$$\frac{\alpha_1}{\alpha_2} = \frac{c_1^t}{c_2^t}$$

Now, consider the case when the agent is solving the problem, agent is representative agent, the agent is will be solving for the given time-o price at t^{th} time-period as P_t , Agent, problem is as follows:-

$$\begin{aligned}
&\max_{\{C_i^t\}_{t=0}^{\infty}} \sum_{t=0}^{\infty} \beta^t u(C_i^t) \\
&\text{set. } \sum_{t=0}^{\infty} P_t C_t^i = \sum_{t=0}^{\infty} P_t y_t^i + P_0 a_0^i
\end{aligned}$$

Setting up the Lagrangian, there will be only one constraint as there is

only one time-0 constraint.

$$\mathcal{L} = \sum_{t=0}^{\infty} \beta^t u(C_i^t) + \lambda_i \left[\sum_{t=0}^{\infty} P_t y_t^i - \sum_{t=0}^{\infty} P_t C_t^i + P_0 a_0^i \right]$$

Solving for the FOCs

$$\beta^t u'(c_i^t) + \lambda_i (-P_t) = 0 \quad \forall t \quad \forall i$$

Therefore, for both of the agents,,

$$\beta^t u'(c_1^t) = \lambda_1 P_t \quad \forall t$$

$$\beta^t u'(C_2^t) = \lambda_2 P_t \quad \forall t$$

The ratio of the marginal utilities

$$\frac{u'(c_1^t)}{u'(c_2^t)} = \frac{\lambda_1}{\lambda_2} \quad \forall t$$

Let us consider the case for simplicity that the utility function is logarithm function, $u = \log$:

$$\frac{\beta^t}{c_1^t} = \lambda_1 P_t$$

$$c_1^t = \frac{\beta^t}{\lambda_1} P_t \quad \forall t$$

$$\sum_{t=0}^{\infty} P_t \frac{\beta^t}{\lambda_i P_t} = \sum_{t=0}^{\infty} P_t y_t^i + P_0 a_0^i$$

$$\frac{1}{\lambda_i} \sum_{t=0}^{\infty} \beta^t = \sum_{t=0}^{\infty} p_t y_t^i + P_0 \partial_0^i \quad \forall i$$

$$1/\lambda_i = (1 - \beta) \left[\sum_{t=0}^{\infty} P_t y_t^i + P_0 \partial_0^i \right] \quad \forall i$$

$$\Rightarrow c_i^t = \beta^t \times \frac{1}{(1 - \beta) [\sum_t P_t y_t^i + P_0 a_0^i]} P_t$$

$\forall i$

Mapping with planner's problem $[\alpha_1 = \alpha, \alpha_2 = 1 - \alpha]$

planner's problem $[\alpha_1 = \alpha, \alpha_2 = 1 - \alpha]$

$$\frac{\sum_t P_t y_t + P_0 a_0^2}{\sum_t P_t y_t + P_0 a_0^1} = \frac{1 - \alpha}{\alpha} \Rightarrow \alpha = \frac{\sum_1 P_t y_t + P_0 a_0^i}{2 \sum_t P_t y_t + P_0 (a_0^i + \alpha_0^2)}$$

$$\Rightarrow \alpha = \frac{\sum p_t y_t + \alpha_0^i}{2 \sum_t p_t y_t}$$

Prices: $\frac{P_{t+1}}{P_t} = \frac{\theta_{t+1}}{\theta_t}$
 (from Plane's problem)

$$\Rightarrow \frac{P_1}{P_0} = \frac{\theta_1}{\theta_0}$$

$$\text{let } P_0 = \theta_0 = 1$$

$$\Rightarrow P_t = \theta_t$$

$\forall t.$

- Sequential markets Consumer (RA) problem

$$\max_{\{c_t^i\}_{t=0}^{\infty}} \sum_{t=0}^{\infty} \beta^t u(c_t^i)$$

subject to: $q_t a_{t+1}^i + c_t^i = y + \partial_t^i \quad \forall t$

$$\partial_0^i \geq 0$$

and $\lim_{T \rightarrow \infty} \frac{q_T a_{T+1}^i}{\prod_{s=1}^T (1+r_s)} \geq 0$

here. $r_{t+1} = \frac{1}{q_t} - 1$

in Recursive form:-

$$V(\partial_i, A_i) = \max_{\partial_i', c_i} \{u(c_i) + \beta v(\partial_i', A_i')\}$$

Subject to

$$q(A_i) a_i' + c_i' = y + a_i$$

$$A_i' = G(A_i)$$

FOL $\frac{\partial v(\partial_i, A_i)}{\partial \partial_i'} = 0 \Rightarrow 0 = u'(y - \partial_i - q(A_i) \partial_i') [-q(A_i)] + \frac{\partial v(\partial_i', A_i')}{\partial \partial_i'}$
 from envelope theorem, we know

$$\frac{\partial v(\partial_i', A_i')}{\partial \partial_i'} = u'(y - \partial_i' - q(A_i') \partial_i'')$$

Substituting, we get

$$u'(y - \partial_i - q(A_i) \partial_i') = \beta (1 + r(A_i)) u'(y - \partial_i' - q(A_i') \partial_i'')$$

this is the Euler equation. $\left(1 + r(A_i) = \frac{1}{g(A_i)}\right)$
 CEq^m with Heterogeneity: (Recursive)

A RCE is a value function $v(\partial_i, A_1)$, a decision rule $g(a_i, A_1)$ and a price function $q(A_1)$, such that:

1. g solves the consumer's problem given G and q . v is the associated value function.
2. Consistency:

$$G(a_1) = g(a_1, a_1)$$

3. Market clearing:

$$g(a_1, a_1) + g(-a_1, a_1) = 0$$

3 Uncertainty

3.1 *Introduction:*

In the real world scenarios, we do not observe the state of the world in future period with certainty. Therefore, models under certainty has a limited role in explaining the real problem in the economy. This uncertainty is primarily captured through the concept of the state space, which represents all possible conditions or states of the economy at a given time. These states are described by variables known as state variables. When these state variables are subject to randomness or variability, we use probability theory to model how consumers and the economy behave.

The first topic we'll delve into is the exploration of common stochastic processes in macroeconomics. Stochastic processes are mathematical models that describe the evolution of variables over time in a probabilistic manner. These processes help us understand how economic variables change and interact under uncertainty.

Secondly, we'll examine how individuals make decisions to maximize their utility when faced with uncertainty. Since outcomes are not certain, individuals must make choices based on probabilities and expected outcomes to maximize their satisfaction or utility.

Lastly, we'll define and discuss competitive equilibrium in the context of uncertainty. Competitive equilibrium refers to a state where markets are in balance, with supply equaling demand across all goods and services. We'll explore how this equilibrium is affected by uncertainty and its implications for the functioning of the economy. Before proceeding to these topic we will first define some basic concepts in stochastic process, these concepts are in the Appendix11.

3.2 *Markov Chains:*

Consider a stochastic process describing weather dynamics in a fixed geographic location (e.g., Montreal). Let

$$\{Z_t\}_{t \geq 0}$$

be a sequence of random variables, where each Z_t takes values in a finite state space Z , representing possible weather types (for example,

$Z = \{w, s\}$ for winter and summer). A realization of the process is a sequence such as

$$w, w, s, s, s, s, w, w, \dots$$

The evolution of the process is governed by conditional probabilities of the form

$$\Pr(Z_{t+1} = z' \mid Z_t = z),$$

which specify the probability distribution of tomorrow's weather state given today's state. These probabilities do not depend on the full past history

$$(Z_{t-1}, Z_{t-2}, \dots),$$

but only on the current state Z_t .

Since the initial state of the process is not observed (e.g., the weather state at the origin of time), it is characterized by an initial probability distribution over the state space Z . Together with the conditional transition probabilities, this initial distribution fully determines the joint distribution of the process at all future dates.

The stochastic process described above is a Markov process. In discrete time with a finite state space, it is referred to as a Markov chain.

Formally, a Markov chain $\{Z_t\}_{t \geq 0}$ is defined by:

1. A finite state space

$$Z = \{\bar{z}_1, \bar{z}_2, \dots, \bar{z}_n\}.$$

2. A transition matrix $P \in \mathbb{R}^{n \times n}$, where

$$P_{ij} = \Pr(Z_{t+1} = \bar{z}_j \mid Z_t = \bar{z}_i), \quad \sum_{j=1}^n P_{ij} = 1 \quad \forall i.$$

3. An initial distribution $\pi \in \mathbb{R}^{1 \times n}$, where

$$\pi_j = \Pr(Z_0 = \bar{z}_j), \quad \sum_{j=1}^n \pi_j = 1.$$

Given (π, P) , the entire stochastic process $\{Z_t\}_{t \geq 0}$ is fully characterized.

Given the one-period transition probabilities and an unconditional distribution over states, we can define multi-period transition probabilities. In particular, the two-period transition probability from state $\bar{z}_i$

to state $\bar{z}_j$ is given by

$$\Pr(Z_{t+2} = \bar{z}_j \mid Z_t = \bar{z}_i) = \sum_{h=1}^n \Pr(Z_{t+2} = \bar{z}_j \mid Z_{t+1} = \bar{z}_h) \Pr(Z_{t+1} = \bar{z}_h \mid Z_t = \bar{z}_i) = \sum_{h=1}^n P_{ih} P_{hj}.$$

In matrix notation, this can be written as

$$\Pr(Z_{t+2} = \bar{z}_j \mid Z_t = \bar{z}_i) = (P^2)_{ij}.$$

More generally, the k -period transition probabilities are given by the k -th power of the transition matrix:

$$\Pr(Z_{t+k} = \bar{z}_j \mid Z_t = \bar{z}_i) = (P^k)_{ij}.$$

Let $\pi_t \in \mathbb{R}^{1 \times n}$ denote the unconditional distribution of Z_t . The law of motion for the unconditional distribution is

$$\pi_{t+1} = \pi_t P,$$

so that

$$\pi_t = \pi_0 P^t \quad \text{for all } t \geq 0.$$

A distribution π is said to be a *stationary distribution* if it is invariant under the transition matrix, that is,

$$\pi = \pi P,$$

or equivalently,

$$\pi(I_n - P) = 0,$$

subject to the probability constraints $\pi_j \geq 0$ for all j and $\sum_{j=1}^n \pi_j = 1$.

The stationary distribution, when it exists, represents a situation in which the unconditional distribution of states remains unchanged over time.

From the invariance condition $\pi = \pi P$, it follows that π is a left eigenvector of the transition matrix P associated with the eigenvalue 1. Since P is a row-stochastic matrix, 1 is always an eigenvalue of P ⁵.

⁵An eigenvector v of a matrix P associated with eigenvalue λ satisfies

$$vP = \lambda v.$$

When $\lambda = 1$, this condition implies that applying the transition matrix P leaves the vector unchanged. In the context of Markov chains, a stationary distribution is a probability vector that is invariant under the transition dynamics, and therefore corresponds to a left eigenvector of P with eigenvalue 1.

The existence and uniqueness of the stationary distribution depend on the properties of the Markov chain. In particular:

- A stationary distribution exists for any finite Markov chain.
- The stationary distribution is unique if the Markov chain is irreducible and aperiodic.

Under these conditions, $\pi_t \rightarrow \pi$ as $t \rightarrow \infty$, regardless of the initial distribution π_0 .

When the state variable evolves according to a continuous-state Markov process, such as an AR(1), it is often convenient to approximate it by a finite-state Markov chain. Consider the AR(1) process

$$z_t = \rho z_{t-1} + \sigma_\varepsilon \varepsilon_t, \quad \varepsilon_t \sim \text{i.i.d. } \mathcal{N}(0, 1),$$

with unconditional variance

$$\text{Var}(z_t) \equiv \sigma_z^2 = \frac{\sigma_\varepsilon^2}{1 - \rho^2}.$$

To approximate this continuous-state process by a Markov chain, three elements are required:

1. A finite state space $Z = \{\bar{z}_1, \dots, \bar{z}_n\}$, obtained by discretizing the continuous support of z_t .
2. A transition matrix $P \in \mathbb{R}^{n \times n}$, where

$$P_{ij} = \Pr(z_{t+1} = \bar{z}_j \mid z_t = \bar{z}_i),$$

constructed to approximate the conditional distribution implied by the AR(1) process.

3. A criterion for accuracy, such as the ability of the discretized process to match moments, persistence, or the conditional distribution of the original continuous process.

Early and influential methods for constructing such approximations were developed by Tauchen (1986), with later refinements by Rouwenhorst (1995). These approaches differ in how the state space is constructed and how transition probabilities are assigned, but both aim to preserve key properties of the underlying AR(1) process, such as its unconditional variance and autocorrelation.

Details of these discretization methods are provided in Appendix 12.

3.3 *Neo-classical Growth Model with Uncertainty*

We now study the planner's problem under uncertainty using a sequence formulation. Consider a social planner operating a technology

$$y_t = z_t F(k_t, n_t),$$

where productivity z_t is stochastic. For simplicity, we assume labor is inelastic and normalized to one, $n_t = 1$. (Models with endogenous labor choice are typically referred to as real business cycle (RBC) models.)

Timing within period t is as follows. The capital stock k_t is pre-determined at the beginning of period t . The productivity shock z_t is realized, after which consumption c_t and next period's capital k_{t+1} are chosen.

We assume that productivity follows a stochastic process. For now, we restrict attention to the case in which $\{z_t\}$ is a finite-state Markov chain with strictly positive support, $\bar{z}_i > 0$ for all i .

Let the history of productivity realizations up to time t be denoted by

$$z^t = (z_t, z_{t-1}, \dots, z_0) = (z_t, z^{t-1}) \in Z^{t+1},$$

where superscripts denote histories and subscripts denote realizations at a given date.

Let $\pi(z^t) > 0$ denote the unconditional probability of history z^t . Conditional probabilities are defined by

$$\pi(z_{t+1} \mid z^t) = \frac{\pi(z_{t+1}, z^t)}{\pi(z^t)}.$$

The planner chooses state-contingent sequences $\{c(z^t), k(z^t)\}_{t \geq 0}$ to solve

$$\max_{\{c(z^t), k(z^t)\}_{t \geq 0}} \sum_{t=0}^{\infty} \sum_{z^t \in Z^{t+1}} \beta^t \pi(z^t) u(c(z^t))$$

subject to the resource constraints

$$c(z^t) + k(z^t) = z_t F(k(z^{t-1}), 1) + (1 - \delta)k(z^{t-1}), \quad \forall t \geq 0, \quad z^t \in Z^{t+1},$$

and the feasibility conditions

$$c(z^t) \geq 0, \quad k(z^t) \geq 0,$$

with initial capital $k_0 > 0$ given.

The first-order conditions characterize the optimal allocation and

lead to a stochastic Euler equation linking marginal utility across histories.

The planner chooses state-contingent sequences $\{c(z^t), k(z^t)\}_{t \geq 0}$ to maximize expected discounted utility subject to the resource constraints. The first-order condition with respect to capital yields the stochastic Euler equation:

$$u'(c(z^t)) = \beta \sum_{z_{t+1} \in Z} \pi(z_{t+1} | z_t) u'(c(z^{t+1})) \left[z_{t+1} F_k(k(z^t), 1) + (1 - \delta) \right].$$

This condition equates the marginal utility cost of foregoing one unit of consumption today to the expected discounted marginal utility benefit tomorrow. The benefit consists of two components: the marginal product of capital in period $t + 1$, which depends on the realized productivity shock z_{t+1} , and the undepreciated portion of capital.

The period-by-period resource constraint is

$$c(z^t) + k(z^t) = z_t F(k(z^{t-1}), 1) + (1 - \delta)k(z^{t-1}),$$

with feasibility conditions $c(z^t) \geq 0$, $k(z^t) \geq 0$, and initial capital $k_0 > 0$ given.

Optimality also requires the transversality condition

$$\lim_{t \rightarrow \infty} \sum_{z^t \in Z^{t+1}} \pi(z^t) \beta^t u'(c(z^t)) k(z^t) = 0,$$

which rules out plans that accumulate capital indefinitely without providing utility benefits.

Recursive Formulation

Because preferences are time-separable and productivity follows a Markov process, the planner's problem admits a recursive representation. Let $v(k, z)$ denote the value function when the current state is (k, z) . The Bellman equation is

$$v(k, z) = \max_{k' \in [0, zF(k, 1) + (1 - \delta)k]} \left\{ u(zF(k, 1) + (1 - \delta)k - k') + \beta \sum_{z' \in Z} \pi(z' | z) v(k', z') \right\}.$$

The optimal policy function $k' = g(k, z)$ maps current states into next-period capital choices. The Euler equation derived from this recursive problem coincides with the sequence-form Euler equation.

In the special case of logarithmic utility, Cobb–Douglas production,

and full depreciation ($\delta = 1$), the policy function takes the closed-form solution

$$k' = \alpha\beta z k^\alpha.$$

Dynamics and Distribution of Capital

Given the policy function g and the stochastic process for productivity, the evolution of capital becomes stochastic. If $\{z_t\}$ is i.i.d. with cumulative distribution function H , then

$$k_{t+1} = \alpha\beta z_t k_t^\alpha.$$

Starting from an initial capital stock k_0 , the induced one-step-ahead distribution of capital is

$$\psi_1(a) = \Pr(k_1 \leq a) = \Pr\left(z_0 \leq \frac{a}{\alpha\beta k_0^\alpha}\right) = H\left(\frac{a}{\alpha\beta k_0^\alpha}\right).$$

This defines the transition function

$$P(a, b) = \Pr(k_{t+1} \leq a \mid k_t = b) = H\left(\frac{a}{\alpha\beta b^\alpha}\right).$$

Finally, the law of motion for the distribution of capital ψ_t is given by

$$\psi_{t+1}(a) = \Pr(k_{t+1} \leq a) = \int P(a, b) d\psi_t(b).$$

Thus, even though individual capital choices follow a simple policy rule, aggregate dynamics are described by the evolution of a distribution over capital.

Stationary Distribution and Convergence

Existence and Uniqueness of the Stationary Distribution. Let $\{\psi_t\}_{t \geq 0}$ denote the sequence of probability distributions over capital induced by the policy function g and the shock process. Define the transition operator $\mathcal{T}$ on the space of probability measures by

$$(\mathcal{T}\psi)(A) = \int P(A \mid k) d\psi(k),$$

for any Borel set $A \subset \mathbb{R}_+$, where $P(\cdot \mid k)$ is the transition kernel induced by

$$k_{t+1} = g(k_t, z_t).$$

A stationary distribution ψ^* is a fixed point of this operator:

$$\psi^* = \mathcal{T}\psi^*.$$

Under standard conditions—namely continuity of g , boundedness of moments, and irreducibility of the induced Markov process—the operator $\mathcal{T}$ admits a unique fixed point. Hence, there exists a unique stationary distribution for capital.

Asymptotic Stationarity. For any initial distribution ψ_0 , the sequence $\{\psi_t\}$ defined by

$$\psi_{t+1} = \mathcal{T}\psi_t$$

converges weakly to the stationary distribution ψ^* :

$$\psi_t \Rightarrow \psi^* \quad \text{as } t \rightarrow \infty.$$

This follows from the fact that the induced Markov process is ergodic⁶, implying that long-run outcomes are independent of initial conditions.

An example

Suppose productivity shocks satisfy

$$\ln z_t \sim \mathcal{N}(\mu, \sigma^2),$$

and consider the policy rule

$$k_{t+1} = \alpha\beta z_t k_t^\alpha.$$

Taking logs yields

$$\ln k_{t+1} = \ln(\alpha\beta) + \alpha \ln k_t + \ln z_t.$$

This is an AR(1) process in $\ln k_t$ with autoregressive coefficient $\alpha \in (0, 1)$.

Standard results for AR(1) processes imply that $\{\ln k_t\}$ converges in

⁶In a finite-state Markov chain, ergodicity is equivalent to the chain being irreducible and aperiodic. Irreducibility means every state can be reached from every other state with positive probability in some finite number of steps. The stationary distribution assigns positive probability to every state in an irreducible chain. So, in an informal sense, an ergodic process eventually explores the entire state space, but not uniformly and not at every horizon.

distribution to a normal random variable with unconditional moments

$$\lim_{t \rightarrow \infty} \mathbb{E}_0[\ln k_t] = \frac{\ln(\alpha\beta) + \mu}{1 - \alpha},$$

$$\lim_{t \rightarrow \infty} \mathbb{V}_0[\ln k_t] = \frac{\sigma^2}{1 - \alpha^2}.$$

The above expressions are very clean and easy to implement because in this case, the solution is closed-form.

Numerical Implementation

In most applications, closed-form solutions are unavailable. Stationary distributions are therefore computed numerically, either by iterating on the law of motion

$$\psi_{t+1} = \mathcal{T}\psi_t,$$

until convergence, or by simulating long time series of the model and approximating the invariant distribution using simulated moments.

4 Market Structures

4.1 Introduction

In macroeconomics, market structure refers to the organization and characteristics of markets in which agents trade goods, labor, and financial assets. A central distinction is between *complete* and *incomplete* markets. While households face similar intertemporal optimization problems under both structures, the set of available assets and the associated budget constraints differ substantially.

When agents are exposed to uncertainty—arising from idiosyncratic or aggregate shocks—their primary objective is to smooth consumption over time and across states of the world. This requires saving, borrowing, and trading assets contingent on future realizations of uncertainty. Complete markets provide the richest environment for risk sharing, allowing agents to fully insure against idiosyncratic shocks.

In this section, we focus on a complete market structure with a full menu of Arrow securities. We begin with a sequential market formulation and then study its recursive counterpart. Finally, we discuss the implications of complete markets for risk sharing.

4.2 Sequential Markets with Arrow Securities

Consider an infinite-horizon economy with uncertainty represented by histories $z^t \in Z^{t+1}$. Markets are sequential, and at each date and history agents can trade a complete set of Arrow securities. An Arrow security purchased at history z^t pays one unit of consumption in period $t + 1$ if state z_{t+1} is realized.

Let $a(z^t, z_{t+1})$ denote holdings of Arrow securities purchased at z^t , and let $q(z^t, z_{t+1})$ denote their prices. The consumer's problem is to choose state-contingent plans $\{c(z^t), n(z^t), k(z^t), a(z^t, z_{t+1})\}$ to maximize expected utility subject to the sequence of budget constraints:

$$c(z^t) + \sum_{z_{t+1} \in Z} q(z^t, z_{t+1}) a(z^t, z_{t+1}) + k(z^t) \leq a(z^{t-1}, z_t) + (R(z^t) + 1 - \delta)k(z^{t-1}) + w(z^t)n(z^t),$$

for all $t \geq 0$ and all $z^t \in Z^{t+1}$.

The remaining constraints are:

$$c(z^t), k(z^t) \geq 0, \quad \ell(z^t) + n(z^t) = 1, \quad a(z_0) = 0 \text{ for all } z_0 \in Z,$$

with initial capital $k_0 > 0$ given. A no-Ponzi condition ensures solvency:

$$\lim_{t \rightarrow \infty} \sum_{z_{t+1} \in Z} p(z^t) q(z^t, z_{t+1}) a(z^t, z_{t+1}) \geq 0, \quad \text{for all } z^\infty.$$

Definition 2. *Competitive equilibrium with sequential markets consists of:*

- *A price system*

$$p \equiv \{p(z^t), w(z^t), R(z^t), \{q(z^t, z_{t+1})\}_{z_{t+1} \in Z}\}_{t \geq 0},$$

- *Allocations for consumers and firms,*

such that:

1. *Consumers and firms optimize given prices;*
2. *Markets clear:*

$$k^d(z^t) = k^s(z^{t-1}),$$

$$n^d(z^t) = n^s(z^t),$$

$$c(z^t) + i(z^t) = y(z^t),$$

$$a(z^t, z_{t+1}) = 0 \text{ for all } z_{t+1} \in Z.$$

Recursive Formulation

Given Markov shocks and time-separable preferences, the problem admits a recursive representation. Let η denote financial wealth and (K, z) aggregate states. The consumer's value function solves:

$$v(\eta, z, K) = \max_{c, n, k', \{a'(z')\}} \left\{ u(c, 1 - n) + \beta \sum_{z' \in Z} \pi(z' | z) v(\eta', z', K') \right\},$$

subject to

$$c + \sum_{z' \in Z} q(z, K; z') a'(z') + k' - w(z, K) n \leq \eta,$$

$$\eta' = a'(z') + (R(z', K') + 1 - \delta) k', \quad K' = G_K(z, K), \quad N = G_N(z, K).$$

4.3 *Implications of Complete Markets: Risk Sharing*

The first-order condition with respect to Arrow securities implies:

$$q(z, K; z') = \pi(z' | z) \beta \frac{u_c(c', 1 - n')}{u_c(c, 1 - n)} = \pi(z' | z) \text{MRS}_{c', c}.$$

With a risk-free bond, its price satisfies:

$$q_b(z, K) = \sum_{z' \in Z} \pi(z' | z) \beta \frac{u_c(c', 1 - n')}{u_c(c, 1 - n)}.$$

Thus, the bond is redundant in the presence of a complete set of Arrow securities.

For two individuals i and j , complete markets imply:

$$\text{MRS}_{t+1, t}^i(z') = \text{MRS}_{t+1, t}^j(z') \quad \forall z',$$

which is stronger than the condition obtained with only a risk-free bond:

$$\mathbb{E}_t \text{MRS}_{t+1, t}^i = \mathbb{E}_t \text{MRS}_{t+1, t}^j.$$

With homothetic preferences, this implies:

$$\frac{c_{t+1}^i}{c_t^i} = \frac{c_{t+1}^j}{c_t^j},$$

so individual consumptions move proportionally with aggregate consumption. Idiosyncratic shocks do not affect consumption growth.

A more detailed example is discussed at the end Section: 13

4.3.1 *Endowment Economy Illustration*

Consider an economy with I agents facing idiosyncratic endowment shocks $y_i(s^t)$. The efficient allocation solves:

$$\max_{\{c_i(s^t)\}} \sum_{i=1}^I \lambda_i \sum_{t=0}^{\infty} \sum_{s^t} \beta^t \pi(s^t) u(c_i(s^t))$$

subject to

$$\sum_{i=1}^I c_i(s^t) = \sum_{i=1}^I y_i(s^t), \quad \forall t, s^t.$$

The optimal allocation satisfies:

$$\frac{u'(c_i(s^t))}{u'(c_j(s^t))} = \frac{\lambda_j}{\lambda_i},$$

implying

$$c_i(s^t) = \lambda_i \sum_{j=1}^I y_j(s^t).$$

Thus, individual consumption depends only on aggregate endowment:

1. It is history-independent;
2. It is independent of the cross-sectional distribution of endowments.

A property of this solution under a special case in which the weights are same for everyone, we studied this in our Micro-1 graduate course.

In a symmetric equilibrium ($\lambda_i = \lambda$) with no aggregate uncertainty,

$$c_i(s^t) = \frac{\bar{y}}{I}, \quad \forall i, t.$$

With logarithmic preferences,

$$c_i(s^t) = \lambda_i \sum_{j=1}^I y_j(s^t),$$

implying

$$\frac{c_i(s^{t+1})}{c_i(s^t)} = \frac{\sum_j y_j(s^{t+1})}{\sum_j y_j(s^t)} = \frac{\sum_j c_j(s^{t+1})}{\sum_j c_j(s^t)}.$$

5 Asset pricing model

5.1 *Lucas Asset Pricing Model*

5.2 *Equity Premium Puzzle*

In this complete market asset pricing model, consider CRRA preferences, z (aggregate state) takes good (g) or bad (b). The consumer has access to three assets: two arrow securities (with price Q_z and quantity a_z), and a tree that lasts only for one period and the consumer receives one unit of a new tree every period as an endowment (tree price for quantity x at p_z). The problem of the representative consumer is to solve

$$V(a, x, z) = \max_{a'_g, a'_b, x'} \frac{c^{1-\sigma}}{1-\sigma} + \beta \sum_{z'=g,b} \phi_{zz'} V(a', x', z')$$

s.t.

$$c + Q_g a'_g + Q_b a'_b + p_z(x' - 1) = a + xY_z$$

6 Incomplete Markets

Incomplete markets refer to situations where Arrow securities, which represent complete sets of securities for all possible states or shocks, are not available. In our previous studies, we focused on models with representative consumers, where the behavior of macroeconomic outcomes didn't rely on individual differences. We examined cases where shocks affected aggregate outcomes. Now, we'll delve into modeling scenarios where individuals experience idiosyncratic shocks. These shocks are not insurable due to the absence of complete markets.

Studying incomplete markets provides insights into the origins of wealth inequality, wealth mobility, and the consequences of financial market development. In the planner's problem, we observed that individual consumption solely relies on the aggregate endowment. When the aggregate endowment remains constant across different states, consumption becomes fully insured over time and across states for each individual. Mace (1991) tested this model, investigating whether, under perfect risk-sharing and Constant Relative Risk Aversion (CRRA) preferences, individual consumption should react to aggregate income shocks but not to changes in individual income. That is in the model, $\Delta \log c_t^i = \alpha_1 \Delta \log C_t + \alpha_2 \Delta \log y_t^i + \epsilon_t^i$ testing under the complete market is $H_0 : \alpha_1 = 1$ and $\alpha_2 = 0$ and testing under Autarky is $H_0 : \alpha_1 = 0$ and $\alpha_2 = 1$. Both of the hypothesis were rejected.

In the Lucas asset pricing model, we derived the asset prices and model the general equilibrium framework, whereas, now we will explore the partial equilibrium with prices as given. Therefore, now we refer to situations where certain risks or shocks cannot be fully mitigated through the purchase of insurance. In such markets, individuals cannot easily transfer the risk associated with adverse events to others, as there may not be available markets for insurance against these specific shocks.

Agents in incomplete markets must resort to "self-insurance" strategies to manage the impact of adverse shocks on their consumption. This involves managing a stock of a single asset, such as savings or investments, to cushion the effects of income fluctuations or other adverse events on their consumption levels, also known as consumption smoothing.

The effectiveness of self-insurance strategies depends on several factors:

1. **Utility Function:** The shape and properties of an individual's utility function play a crucial role in determining their willingness to sacrifice current consumption to build up savings or investments for future consumption smoothing. Different utility functions may lead to varying degrees of risk aversion and preferences for consumption smoothing.
2. **Income Process:** The nature of the income process, including its volatility, persistence, and correlation with other economic variables, influences the degree to which individuals need to self-insure. Higher income volatility or persistence of shocks may require individuals to hold larger buffers of assets to smooth consumption over time.
3. **Existence of Borrowing Constraints:** Borrowing constraints restrict an individual's ability to borrow against future income or assets, limiting their capacity to smooth consumption in the face of adverse shocks. Incomplete markets often coincide with the presence of borrowing constraints, which further complicate self-insurance strategies.

6.1 *Income Fluctuation Problem*

We will start modeling the income fluctuation problem in the basic incomplete market setting. Consider an endowment economy, where endowment $y_t \in \mathcal{Y} = \{y_1, y_2, \dots, y_S\}$ follows a Markov chain with $\pi(y'|y) = \Pr(y_{t+1} = y' | y_t = y)$ and $\Pi(y) = \Pr(y_t = y)$ be the conditional and unconditional probability distribution of the state. We will consider the case when only one asset is available, risk-free bond and idiosyncratic element in endowment shocks. Given this, agents would like to self-insure, falling short of full insurance. There is a continuum of individuals of measure 1, and therefore, the the proportion of individuals being in a particular state is same as the probability distribution over $[0,1]$. There is no aggregate uncertainty, and therefore,

$$\bar{y} = \sum_y y \Pi(y)$$

Individual's type at time t is identified by state space.

Let us write the consumer problem, first, state variables are total wealth, income with the individual ω and also the endowment at time t . Since, the problem is stationary here, t would not be a state variable. In the recursive formulation, consumer problem is as follows:

$$V(\omega, y) = \max_{a' \geq -\phi} \{u(w - a') + \beta \sum_{y'} \pi(y'|y) V(w', y')\}$$

sub. to

$$\begin{aligned} c + a' &\leq \omega \\ \omega' &\equiv y' + (1 + r)a' \end{aligned}$$

Here, ω is cash-on-hand. If shocks are persistent⁷, there is no difference in taking a as a state variable in replace of the ω . Note that here the constraint on a' in the maximisation is credit constraints.

Consider the case of no uncertainty, and $\{y_t\}$ follows a deterministic path, or is a deterministic sequence of endowment. This will make 2 changes in the recursive formulation mentioned above, 1) time comes in as a state variable, and 2) no probability. One more condition or constraint is nPg (non-ponzi game), which requires agent should not die with debt, this requires that the agent at any time t can not borrow more than the life time income. Mathematically,

$$a_{t+1} \geq \sum_{\tau=0}^{\infty} \frac{c_{t+\tau}}{(1+r)^\tau} - \sum_{\tau=0}^{\infty} \frac{y_{t+\tau}}{(1+r)^\tau}$$

For the upper bound on the borrowing, imposing $c_t = 0 \forall t$ we get,

$$a_{t+1} \geq - \sum_{\tau=0}^{\infty} \frac{y_{t+\tau}}{(1+r)^\tau} \equiv -\hat{b}_t$$

Present value budget constraint will determine the maximal feasible for an agent to pay in every period forever.

$$\begin{aligned} PV &= \sum_{t=0}^{\infty} \frac{x}{(1+r)^t} \\ PV &= \frac{x}{r} \\ x &= PV.r \end{aligned}$$

Here, PV is $\hat{b}_t$ at time t . therefore, annuity value of $\{y_t\}$ from t onwards is $r\hat{b}_t$. The natural borrowing constraint is the one which is least stringent, that satisfies the nPg condition. It is denoted by $\hat{b} = \sup_t \hat{b}_t$.

⁷A shock is persistent if its effect take longer to dissipate than the length of the time step. Mathematically, z is persistent if $p(z'|z) \neq p(z')$

We can definitely make this more tight, by increasing it by some small number. Note that it is same as imposing the nPg condition, it does not bind in the finite time as strictly positive consumption is optimal. By setting the borrowing constraint as 0 implies autarky, and there is some interval in between 0 and $\hat{b}$, these borrowing constraint (b) are known as ad-hoc and is potentially binding that is binding sometimes and does not bind sometime. We consider the borrowing limit as a function

$$\phi = \min\{b, \hat{b}\}$$

Coming back to the uncertainty, the worst case scenario for any individual is to receive minimum endowment throughout the life time. Consider $\inf y_t = y_1$, this gives us the natural borrowing limit as

$$a_{t+1} \geq - \sum_{\tau=0}^{\infty} \frac{y_1}{(1+r)^\tau} \equiv - \frac{y_1}{r}$$

and this gives us the $\phi = \min\{b, \frac{y_1}{r}\}$, here b is the ad-hoc borrowing constraint.

Recall from the complete markets that consumption does not track individual's income, as that becomes the function of aggregate income. Here, in the incomplete market, we will see the consumption tracks the income. Solving the consumer problem under uncertainty, if the constraint $a' > -\phi$, we get the Euler equation as:

$$u'(c) \geq \beta(1+r) \sum_{y'} \pi(y'|y) u'(c')$$

The inequality within the Euler equation suggests that, due to borrowing constraints and incomplete markets, individuals are not at their optimum, and they strive to reach it. It's important to note that the borrowing limit is sometimes restrictive, depending on the income state or y . In contrast to complete markets where consumption is solely determined by aggregate income and individuals are fully insured, in incomplete markets, consumption varies with individual income. Furthermore, because of the occasionally binding borrowing constraints, consumption closely follows individual income, exceeding what can be attributed solely to the absence of certain markets.

Looking into the empirical counterpart, we will go through calibrating the model based on the estimates from other studies. Model cali-

bration refers to the process of adjusting the parameters of a theoretical economic model to match specific empirical data or targets. This involves fine-tuning the model's parameters such as coefficients, elasticities, or other structural features to ensure that the model generates outcomes that closely resemble observed economic phenomena or desired macroeconomic indicators. The goal of calibration is to enhance the model's predictive power and its ability to provide meaningful insights into real-world economic behavior and policy implications.

In our model, the empirical counterpart of $\phi(b)$ for the model calibration is to target empirical counterpart of the ratio of the log deviations in consumption to income i.e, $m(\phi) = \frac{std(\log c_i)}{std(\log y_i)}$. Logarithm just for the ease to interpret in terms of relative change and levels.

The range of $m(\phi)$ explains various models,

1. When $\phi = 0$, this implies no borrowing and thus everyone consumes their own endowment (forced to consume), and is a Autarky condition. This is also known as hand-to-mouth.
2. For the value of $\phi = y_1/r$ that is the natural borrowing limit, this gives us the lower bound on the ϕ , and $m(y_1/r) \in (0, 1)$ this implies that markets are incomplete.
3. If $\phi \in (0, y_1/r)$, we get $m(\phi) \in (m(y_1/r), 1)$, still incomplete, but this cannot be captured by the model, as it goes below the infimum of ϕ .
4. For the case when $m = 0$, we get the condition of complete markets.

Clearly, from the empirical evidence we are far from the complete markets, and therefore the model or our approach works here if $m(y_1/r)$ is lower than empirical counterpart.

6.2 *Huggett Model*

We've covered the partial equilibrium aspect, and now we're transitioning to examine the General Equilibrium analysis of income fluctuations using the Huggett model as outlined in the Journal of Economic Dynamics and Control in 1993.

Consider the state space consist of 1) the individual level (ω, y) and 2)

the aggregate level ψ . ψ captures the overall cross-sectional distribution of (ω, y) , which is an ∞ dimensional object, or simple the motion how individuals move over time. This becomes the aggregate variable as it captures the wealth and income of individual at every given point of time.

Given this, the consumer problem can be written as following,

$$V(\omega, y, \psi) = \max_{a' \geq -\phi} \left\{ u(\omega - a') + \beta \sum_{y'} \pi(y'|y) V(\omega', y', \psi') \right\}$$

s.t.

$$\begin{aligned} c + a' &\leq w \\ w' &\equiv y' + (1 + r(\psi'))a' \end{aligned}$$

with

$$\psi' = H(\psi)$$

Now, we will study the long run equilibrium, but before that let us define some definitions. Here, we will be looking into the subset of the individual state that is economically feasible and interesting to study, that is $(\omega, y) \in X = [\underline{\omega}, +\infty) \times \mathcal{Y}$ with an upper bound on ω . We denote this relevant subset (compact) as S . Let, $\mathcal{B}_S$ be the Borel σ algebra on S . Formally, we say that ψ is a probability measure defined over the space $(S, \mathcal{B}_S)$ ⁸

The transition of function is induced by the $a' = g(\omega, y)$ and π . This we can think of as probability of getting to the state (w', y') given the state today as (w, y) . Formally, $P : S \times \mathcal{B}_S \rightarrow [0, 1]$, with the typical element of it as

$$P((\omega, y), ([\omega_l, \omega_h], \{y_l, \dots, y_h\}) = \sum_{y' \in \{y_l, \dots, y_h\}} \pi(y'|y) \mathbf{1}(\omega' \in [\omega_l, \omega_h]).$$

$\mathbf{1}$ is an indicator function.

Now how the transition happens, the law of motion of ψ for a given

⁸A Borel Sigma algebra includes open sets, closed sets, countable unions, and intersections of these sets, making it a versatile tool for defining measurable sets and constructing functions and measures on the space. In probability theory, the Borel sigma algebra is crucial for defining probability measures on the real line or more general topological spaces. It forms the basis for constructing probability spaces and random variables, allowing us to define events and calculate probabilities in a rigorous manner. Here in our case it is simple as we have the cartesian product of a compact set of $\mathbb{R}$ and finite set.

$B \in \mathcal{B}_S$ is

$$\psi'(B) = \int_S P(x, B) d\psi(x)$$

The intuition behind the above motion equation is integrating (summing) over all the individuals who will land at the state B tomorrow given that they are at the state x today. There can be many individuals who lands there, since we restricted our population to continuum of 1, here the probability integration is just the proportion of the population.

Definition 3. *A Stationary Recursive Competitive Equilibrium (SRCE) is a value function $V(k, K)$, a decision rule, or a policy function $a' = g(\omega, y)$, an interest rate r , an aggregate law of motion ψ over individual states such that:*

1. *Given r , consumer optimizes solving the consumer problem mentioned above representative.*
2. *Market clearing:*

$$\int_S g(\omega, y) d\psi = 0$$

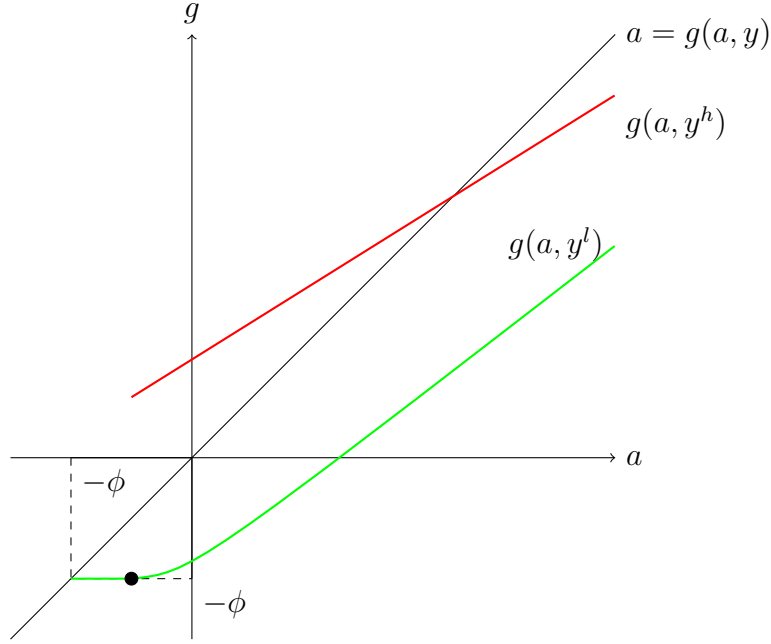
3. *Stationary (Consistency), that is*

$$\psi(B) = \int_S P(x, B) d\psi(x)$$

Where P is induced by π and $g(\omega, y)$

A few things to note till now are, we are studying the workhorse model of the heterogeneous agent model. Mainly focusing on the ex-ante heterogeneity. We treat each agent similar ex-ante, which is described by the state (ω, y) . We are focusing on the case where outcome, interest rate, and distribution ψ are constant in the steady-state. This does not rule out the micro-level action y, a, c . If this would have been a planner's problem the only uncertainty for the planner would be proportion of the individual changing their type, but not how many, as their will be convergence in the distribution and we know the proportion of individual transitioning from one state to another. We will also assume the Ergodic property, which implies that in the long run, the probability that an agent has been all type at least once is positive. In the other way, different individual will go through same experiences in the long-run. Though the short-term dynamics starting from different states can be different. Which becomes the key interest of this model.

Recalling the complete market scenario (for benchmark) the consumption of individual will be $c(y) = \bar{y}$ with the interest rate $r = \beta^{-1} - 1 = \rho$. The resource constraint equation will give us the arrow security prices. From the Monotone mixing condition and the theorem from the SLP, we know there is stationary probability distribution of the distribution.

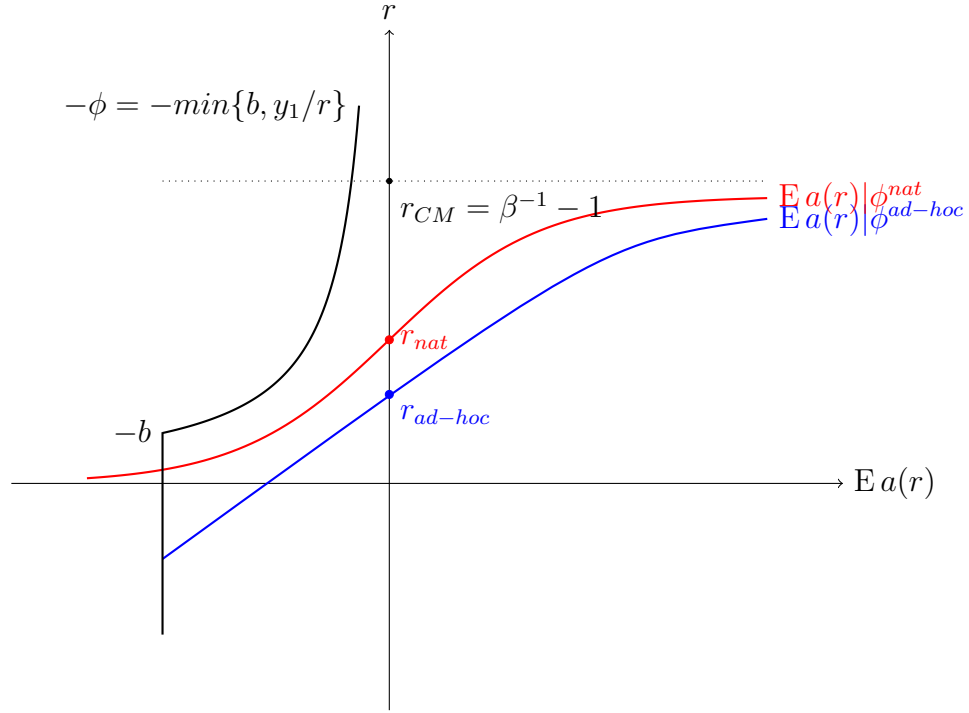


The graph above mentions how the policy function will be for low and high states of the nature. That is for the people who have binding borrowing condition and for whom borrowing conditions does not bind at some state of the world. It is to note that at the bottom of the green graph, the marginal propensity to save is less (next to the dot) and with the increase in the asset the saving increases (i.e., saving increases for those who are far away from them). The marginal propensity to save is the derivative of the curve here. Eventually it becomes parallel to the red line. For whom the marginal propensity to save is constant. It is under the low state of the world, who save less and consume and therefore, the green graph is convex near the black point in the figure. Now we will characterize the equilibrium under the Huggett framework. For the equilibrium, the market should clear and therefore the excess demand of the asset should be 0 as the interest rate reaches the equilibrium interest rate. The expression for the excess demand is

$$E a(r) = \int_S g(\omega, y; r) d\psi_r$$

This implies that as $r \rightarrow \rho$ in the incomplete market $\lim_{r \rightarrow \rho} E a(r) = +\infty$

In the incomplete market the agents are not able to have full insurance and therefore they do some precautionary savings or self-insurance, incomplete markets give them motive to save for consumption smoothing. This gives us that at the complete market interest rate, agents would demand very high amount of the asset to insure themselves. If compensation for time impatience is just right, then the demand for the asset is arbitrarily large, that is if β is low the interest rate is very large. Whereas, if the interest rate $r \rightarrow -1$, the excess demand $E a(r)$ will be negative. People will save for nothing, but to borrow. This characterizes the equilibrium interest rate in incomplete market that equilibrium r will be less than ρ , given continuity.



This is evident from the figure above that the interest rate for the natural borrowing limit is above as compared to the ad-hoc borrowing limit. Tighter the credit constraint lower is the equilibrium interest rate.

Numerical solution: The numerical algorithm to find the equilibrium of the Huggett model involves two loops. One is an inner loop, which constitute of individual optimization, that is taking r as given. Second is an outer loop where we need to iterate over, such that the market clears.

- **Inner loop:** We will first discretize the continuous asset space into $\mathcal{A}$, where the lower bound is $a_1 = -\phi$ and the upper bound as a_{N_a} sufficiently high. We will put smaller intervals at the bottom to capture the curvature.

Then for each $a_i \in \mathcal{A}$ perform value function iteration for the consumer problem (Bellman equation). The consumer problem is

$$v(a_i, y) = \max_{a' \geq -\phi} \{u(y + (1+r)a_i - a') + \beta \sum_{y'} \pi(y'|y) v(a', y')\}$$

Restricting the $a' \in \mathcal{A}$, but more efficient and accurate would be to interpolate $v(a', y')$ (i.e., piece wise linearly) whenever we find $a' \notin \mathcal{A}$. It is better to disregard the credit constraint in the optimization and after the solution is observed, check whether the credit constraint is satisfied. If violated, impose $a' = -\phi$. For efficiency, using the monotonicity property would help, looping over top of $\mathcal{A}$ and then find where, $g(a_j, y) = -\phi$, this gives us that $\forall i < j$ $g(a_j, y) = -\phi$ holds as well.

- **Outer loop:** The main purpose of the outer loop is to find the interest rate r such that the bond market clears. There are two procedures to compute $E a(r)$, the expected excess demand of the asset.

1. A very long simulation of a single agent, iterative jointly on the markov chain and decision rule. Performing piece wise linear interpolation to evaluate $g(a, y)$ at $a \notin \mathcal{A}$.

Discarding the first few observations and approximate $E a(r)$ with the following

$$\frac{1}{T - \hat{T}} \sum_{t=\hat{T}+1}^T a_t$$

The cumulative histogram will yield CDF. For a smoother policy function to have less kinks, solve the bellman equation on finer asset grid. Then simulate using the new decision rule

and alongwith the piece wise linear interpolation whenever $a' \notin \hat{\mathcal{A}}$

2. Another process involves computing the transition function. If $a' \in \mathcal{A}$, then iterate on the discrete pdf ψ , then based on the iteration compute the vector $(1 \times N_a N_y)$ which has a typical element as $\psi(a, y) = Pr((a, y))$. Now, we can compute the transition function which can be arranged in the matrix $\mathbf{P}$ of $(N_a N_y \times N_a N_y)$ dimension, with the typical element as

$$P((a, y), (a', y')) = Pr((a', y')|(a, y)) = \pi(y'|y)\mathbf{1}(g(a, y) = a')$$

For the stationary distribution, this boils down to the problem of fixed point. That is solving the system of linear equations:

$$\begin{aligned}\psi &= \psi \mathbf{P} \\ \psi(\mathbf{I} - \mathbf{P}) &= 0\end{aligned}$$

here, ψ is the eigenvector of $\mathbf{P}$ associated with the unit eigenvalue.

Now, if $a' \notin \mathcal{A}$ then discretizing the density with respect to $\mathcal{A}$, and using the piece wise linear interpolation to split the probabilities. For example, when $a_i < g(a, y) < a_{i+1}$, then,

$$P((a, y), (a_i, y')) = \pi(y'|y) \frac{a_{i+1} - g(a, y)}{a_{i+1} - a_i}$$

$$P((a, y), (a_{i+1}, y')) = \pi(y'|y) \frac{g(a, y) - a_i}{a_{i+1} - a_i}$$

and market clearing condition $\sum_{(a,y)} g(a, y)\psi(a, y) \sim 0$

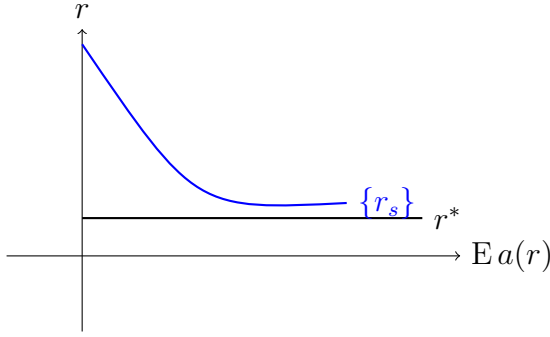
Transitional Dynamics For the case when the distribution ψ_0 is not the steady-state one, then distribution ψ_t will evolve deterministically. The problem is dependent on time and therefore t becomes a state variable. This implies that the value function for the consumer will be the following for any given $t \geq 0$

$$v_t(a, y) = \max_{a' \geq -\phi} \{u(y + (1 + r_t)a_i - a') + \beta \sum_{y'} \pi(y'|y)v_{t+1}(a', y')\}$$

given the perceived law of motion $\{r_s\}_{s=t}^{\infty}$. This gives the policy function

as a function of t and therefore, $a_{t+1} = g_t(a, y)$. Computations for the transitional dynamics, algorithm steps are

1. Guess T , time to complete the transition.
2. Guess a sequence of interest rate, $\{r_s\}_{s=0}^T$ with the initial condition r_0 given. For example, as shown in the figure below, the sequence is converging, the sequence is a perceived law of motion and we guessed and fixed it.
3. Inner loop: Solve the consumer problem by backward induction, that is from $v_{T+1} = v$, where v solves the steady state problem and get the policy function, i.e., $g_t\{a, y\}_{t=0}^T$ and $v_t\{a, y\}_{t=0}^T$



4. outer loop: iterate forward on the economy transition function starting from ψ_0 . This can be done through two ways, one is simulation (i) draw a large number (N) of initial conditions from ψ_0 and then (ii) starting from the $(a_0, y_0)^i$, now simulate for T periods using $\{g_t(a, y)\}_{t=0}^T$ and the Markov chain. second is Iterating on distribution. That is to iterate forward on $\psi_{t+1} = \psi_t P_t$ for T times starting from ψ_0 , using the methods described previously in the second step of the outer loop in the numerical solution of the Huggett model.
5. Now check for the market clearing condition. $N^{-1} \sum_{i=1}^N a_t^i \sim 0$, if not update the sequence of interest rate accordingly.
If not simulating, check whether $r_T \sim r$ or alternatively $\psi_T \sim \psi$. Otherwise increase T .

MIT shocks: Till now, we have seen the transitional dynamics and equilibrium properties. What happens when a surprise shock hits the economy, also known as MIT shock. Example for credit shock as the surprise shock here, that is when ϕ drops. Since, we are looking from the steady state state point of view, the distribution changes, now

the initial distribution is $\psi_0 = \psi^{initial}$ and transition dynamics will also respond and $\psi_t \rightarrow \psi^{final}$. There are two interesting question,

1. What happens in the economy/comparative statics, that is how the aggregate distribution changes?

This can be obtained through comparing the steady states before and after the shock. This is easier to compute but hard to interpret welfare effects. Average welfare loss, in terms of the **consumption equivalent**. For the steady state comparison we try to compute the value of λ_{ss} such that,

$$\int_S E_0 \sum_{t=0}^{\infty} \beta^t u((1-\lambda_{ss})c_t^{initial}(a_0, y_0)) d\psi^{initial} = \int_S v^{initial}(a_0, y_0) d\psi^{final}$$

where, if utility is isoelastic,

$$\lambda_{ss} = \frac{\int_S v^{final}(a_0, y_0) d\psi^{final}}{\int_S v^{initial}(a_0, y_0) d\psi^{initial}} - 1$$

It is very hard to interpret λ_{ss} as it ignores the transitional adjustments, also the distribution has changed due to the shock not only the policy function.

2. What are the welfare effects that captures aggregate and distributional as well?

This is period-0 comparison, a much harder to compute, but meaningful welfare. In period 0, we compute the value of λ_0 such that

$$\int_S E_0 \sum_{t=0}^{\infty} \beta^t u((1-\lambda_0)c_t^{initial}(a_0, y_0)) d\psi^{initial} = \int_S v^{initial}(a_0, y_0) d\psi^{final}$$

More on the MIT shock will be in the General equilibrium framework of incomplete markets, in Aiyagari and Krusell-Smith.

6.3 *Aiyagari Model*

Aiyagri model is an extension of the Huggett model. In this model it has production over and above the Huggett model. With production, consider a representative firm with neo-classical production function ($y = zF(K, N)$). There is no aggregate uncertainty, hence no covariate

shocks. Consumers own's capital and labor. The consumer experience purely idiosyncratic shocks to the labor endowments: $s \in \{0, 1\}$. These shocks are, whether the consumer is employed ($s = 1$) or not employed ($s = 0$). This process is governed by the Markov chain with transition probabilities as $\pi(s'|s)$, with the limiting probability as $\Pi(s)$.

We will be assuming that leisure is not valued, that is it does not enter the consumer's utility function and therefore there is inelastic labor supply, and is also time invariant. We consider the labor supply as $N = \Pi(1)$.

There is no insurance available against the unemployment (that during the negative shock), and there is only one risk free bond. This implies that the markets are incomplete.

The consumer solves the following problem:

$$v(\omega, s) = \max_{k' \geq -\phi} \{u(w - k') + \beta \sum_{s' \in \{0,1\}} \pi(s'|s)v(\omega', s')\}$$

where $\omega' = ws' + (1+r)k'$, we are assuming that total individual saving is k' . The constraint above mentions that the income will be addition of the return from the capital savings and the wage if $s = 1$.

Here the borrowing limit is $\phi = \min\{b, \frac{ws_{min}}{r}\}$. Since, s_{min} is 0, the borrowing constraint is equal to 0, whenever $b \geq 0$.

Once again we will be restricting the state space as a compact subset of $\mathcal{R} \times \{0, 1\}$. Let, κ be a compact subset of $\mathcal{R}_+$ and S be the Cartesian product, i.e, $\kappa \times \{0, 1\}$.

Now we will define the recursive competitive equilibrium in this context.

Definition 4. *A Recursive competitive equilibrium (RCE) is a value function $v(\omega, s)$, a decision rule $k' = g(\omega, s)$, prices r, w , and a probability measure ψ over the individual state space S , such that:*

1. *Consumers optimize given r and w .*
2. *Representative firm optimizes given r and w , i.e,*

$$r = F_K(K, N) - \delta$$

$$w = F_N(K, N)$$

3. *Capital market clears: $\int_S g(\omega, s)d\psi = K$*

4. *Labor market clears: $\Pi(1) = N$*

5. ψ is stationary, and consistent with the individual behavior:

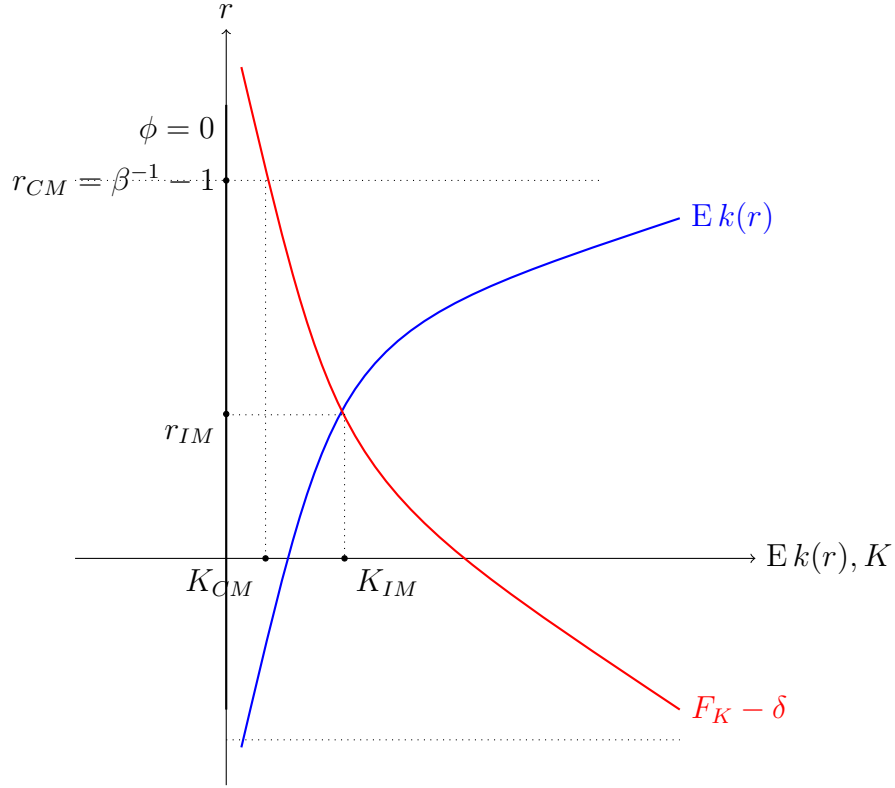
$$\psi(B) = \int_S P(x, B) d\psi(x) \quad \forall B \in \mathcal{BS}$$

Where P is the transition function induced by π and $g(\omega, s)$

As seen in the Huggett model, we will be plotting the aggregate capital, and impose the condition of capital market clearing along with the borrowing constraint of $\psi = 0$. Intuitively, as seen in the Huggett model, the equilibrium interest rate will be much lower than the complete market interest rate. The interest rate from the firm optimization traces the aggregate capital for which $r = F_K(K, N) - \delta$ and therefore is downward slopping, also the aggregate capital demanded by the firm approaches ∞ as the interest rate approaches δ .

Having $\phi = 0$, similar to the Huggett, here at the complete market interest rate the individual's capital supply goes to ∞ . Moreover, as the interest rate approaches the level seen in a complete market, the inter-temporal effect diminishes. However, precautionary savings persist because of market incompleteness, leading individuals to aim for smoother consumption and risk mitigation in the face of uncertainty. Consequently, they supply higher levels of capital to achieve these objectives.

In the Aiyagari model, the equilibrium interest rate is lower compared to a hypothetical scenario where markets are complete, indicating constraints on borrowing. As a result, the capital market clears at a lower interest rate, ensuring that the total amount of capital supplied equals the total amount demanded. However, the equilibrium level of aggregate capital in the economy is higher. This discrepancy between the aggregate savings under incomplete and complete market equilibria ($K^{IM} - K^{CM}$) represents what we call aggregate precautionary savings. Essentially, individuals are supplying more capital and demanding more of the bond. Therefore, in equilibrium individuals save more in anticipation of future income shocks because they cannot fully insure against them due to market incompleteness.



This model highlights the significance of idiosyncratic risk in influencing macroeconomic aggregates. With higher aggregate capital in equilibrium, the overall output of the economy is larger compared to a scenario with complete markets. However, this seemingly positive outcome comes with a caveat. The over-accumulation of capital, driven by precautionary savings to mitigate idiosyncratic risk, leads to inefficiencies. Despite the increase in output, it doesn't necessarily translate to improved welfare for individuals.

Now we will be looking into the what happens when there is aggregate risk involved. Consider productivity shock z in the production function that follows Markov chain. We will be considering business cycle with heterogeneous agents. If s process depends on z that is known as counter cyclical employment risk. We will define the joint transition matrix with typical element $\pi(s', z' | s, z)$.

State variables: Here we still have the state ψ , along with s, z and ω . Now, the consumer problem is as follows:

$$v(\omega, s; \psi, z) = \max_{k' \geq -\phi} \{u(w - k') + \beta \sum_{s', z'} \pi(s', z' | s, z) v(\omega', s'; \psi', z')\}$$

Where, now the wage depends on the next period aggregate capital, next period productivity, and next period Labor force. Which gives

$$\omega' = w(K', N', z')s' + (1 + r(K', N', z'))k'$$

with,

$$\psi' = H(\psi, z, z')$$

$$N = \Pi(s = 1 | z)$$

Note that in the law of motion of population in either of the state is given by H and is also a function of z' which is to find the ψ' .

Now the firm's problem, firm would solve the problem for given r , and w :

$$r = F_K(K, N) - \delta$$

$$w = F_N(K, N)$$

and

$$\psi(B) = \int_S P(x, B) d\psi(x) \quad \forall B \in \mathcal{BS}$$

The definition of the recursive competitive equilibrium remains the same as mentioned above in definition 3, except the change that now we are not in a steady-state with time-invariant ψ . Miao (JET, 2006), Cao (JET, 2020), the theoretical results on the recursive formulation of aggregate state, existence and uniqueness of RCE is non-trivial

6.4 *Krusell-Smith Approach*

The problem outlined above in the Aiyagari model with the aggregate uncertainty poses significant challenges both in numerical and theoretical terms. Krusell and Smith (JPE 1998) proposed an approach to tackle this complexity. They suggested that rather than directly solving for the constraint on r , it suffices to focus on the mean of ψ , as it is a function of ψ . Their investigation delved into the bounded rationality of agents, suggesting that agents can only manage to track a limited number of moments of ψ . They initiated this exploration by considering the first moment, which entails agents tracking only the mean of ψ .

In practice, focusing on the mean for capital, the perceived law of motion is given as:

$$\log K' = a(z) + b(z)\log K, \quad z \in \mathcal{Z}$$

Here, $\log K'$ is the future log of capital (the next period's capital), and $\log K$ is the current log of capital. The parameters $a(z)$ and $b(z)$ depend on the aggregate shock z , which reflects the state of the economy (e.g., productivity shocks, etc.). This simplified law of motion reflects how agents make decisions based on their perception of the economy's aggregate state and their capital.

Perceived Decision Rule

In this framework, individual decision rules can be described as a function $k' = g(k, s; K, z)$, where k' represents future capital, k is current capital, s is any individual-specific state, and K is the mean capital in the economy. The function $g(k, s; K, z)$ governs how individuals choose their savings, given the current capital stock K and the aggregate shock z .

In essence, agents are solving for their optimal capital accumulation, but they do so under the perception that they only need to track the aggregate mean capital, not the entire distribution of capital. This transforms the problem into one more akin to the representative agent model, where the individual decision rule follows a law of motion for capital that depends on the mean level of capital and the aggregate shock.

Implications of the Approach

1. Forecasting Errors

- By focusing on the first moment (the mean), agents may be making forecasting errors because they ignore higher-order moments of the distribution (such as variance or skewness). However, Krusell and Smith showed that these errors are typically small.
- The perceived law of motion for capital is linear in $\log K$, and agents may be wrong about the full distribution of capital, but their errors in forecasting the future capital path are minor.

2. Welfare Implications

- While the approximation introduces forecasting errors, Krusell and Smith argue that these errors do not have large welfare consequences. The reason is that, in most economies, small errors in forecasting the aggregate economy's path (due to focusing on the mean rather than the full distribution) do not result in large misallocations of resources or significant welfare losses.
- Additionally, the mean capital K does a good job of approximating aggregate outcomes, so the welfare cost of ignoring higher moments is small.

3. Approximate Aggregation

- The model features approximate aggregation due to two key properties:
 - **Linear decision rules:** Away from the borrowing constraint, the decision rules for individual savings and capital accumulation are linear. This simplifies the aggregate dynamics and makes it easier to solve the model.
 - **Low marginal propensity to save (MPS) among some agents:** Only a small fraction of agents have a low MPS (for instance, those close to the borrowing constraint). These agents don't contribute much to aggregate savings. Therefore, the overall behavior of the economy is dominated by the more typical agents, who save and accumulate capital at higher rates.

6.4.1 Numerical Algorithm: Krusell Smith

Outer Loop: Iteration on the Law of Motion

The outer loop of the algorithm involves iterating on the law of motion for capital. The objective is to find a fixed point for the parameters $a(z)$ and $b(z)$ for $z \in \mathcal{Z}$, where $\mathcal{Z}$ represents the set of possible aggregate shocks. For example, if the shock set $\mathcal{Z} = \{z_l, z_h\}$ consists of two possible values (say, a low and high shock), then we are solving for four parameters:

$$\{a(z_l), b(z_l), a(z_h), b(z_h)\}.$$

To find the fixed point, the law of motion needs to be consistent with the observed data. Specifically, the model should match the aggregate behavior of the economy. Once the parameters $a(z)$ and $b(z)$ are determined, it is important to check the *goodness of fit*, i.e., verify whether the model's predictions align closely with the simulated data. If the fit is unsatisfactory, adjustments may be necessary. These adjustments could involve allowing agents to track additional moments (such as the variance of capital) or altering the parameterization of the function H that describes the law of motion.

A simple method for computing the fixed point is to simulate a large number of agents over a long time period. The simulated data can then be aggregated to compute the aggregate savings behavior of the economy. Conditional linear regressions are run on the simulated data to estimate the parameters $a(z)$ and $b(z)$.

Inner Loop: Solving the Consumer's Problem

The inner loop involves solving the consumer's optimization problem, given the law of motion for capital. In this step, agents make savings decisions based on their current capital, the state of the economy (represented by the shock z), and their expectations about future capital. With the law of motion for future capital provided by the outer loop, each agent optimizes their savings choice while adhering to their budget constraint and the given law of motion.

A few observations

- **Approximate Aggregation:** Approximate aggregation suggests that if we focus on aggregate variables (e.g., total capital or output), the model simplifies to a neoclassical growth model with aggregate shocks. Individual heterogeneity becomes less important.

This idea is related to Chatterjee (1994), where exact aggregation is achieved.

- **Implications for Insurance:** Approximate aggregation implies that individuals are effectively insured against aggregate shocks. The economy's structure smooths out risks, reducing the exposure of individual agents to aggregate fluctuations.
- **Distributional Concerns:** Although aggregate outcomes can be well approximated, distributional issues remain important. The distribution of wealth or income may still matter for policy and welfare, as highlighted by Chatterjee (1994).

6.5 *Revisiting MIT shocks*

Boppart, Krusell, and Mitman (2018) Approach

Boppart, Krusell, and Mitman (JEDC, 2018) propose a simple method for numerically approximating models via linearization around the steady state. This approach works well when the model exhibits small shocks or minimal non-linearities. In the case of the Aiyagari model with aggregate shocks, this linear approximation is generally valid. If large shocks or significant non-linearities are present, Krusell and Smith's method should be used instead.

MIT Shock Algorithm: Two Main Steps

There are mainly two main steps Deterministic dynamics and Simulation.

1. **Deterministic Dynamics:** Start with the deterministic steady state and compute the Impulse Response Function (IRF) to a one-time surprise Total Factor Productivity (TFP) shock. Let $\log(z_t)$ follow the process:

$$\log(z_t) = \rho \log(z_{t-1}) + \sigma \epsilon_t$$

where ϵ_t is a random shock, typically drawn from a standard normal distribution. For a one-time shock $\epsilon_0 = 1$, the IRF for $\log(z_t)$ will be:

$$\log(z_t) = \sigma, \rho\sigma, \rho^2\sigma, \rho^3\sigma, \dots$$

Next, compute the transitional dynamics of the model, iterating on capital paths $\{K_t\}$ and solving the consumer's problem backward given the future price paths.

For any aggregate variable x , the IRF can be expressed as:

$$\hat{x}(j) \approx \log \left(\frac{x_j}{\bar{x}} \right)$$

where $j \geq 0$ represents the number of periods after the shock.

2. **Simulation:** Simulate the model using random innovations $\{\epsilon_0, \epsilon_1, \dots\}$, with ϵ_t drawn from a standard normal distribution. The outcome at time t is:

$$\hat{x}_t = \sum_{j=0}^t (\epsilon_{t-j} \sigma) \hat{x}(j)$$

The key insight is that the responses to shocks are additive and scalable, which simplifies the analysis of the model.

Intuition from the Representative Agent Model

In the simpler Representative Agent (RA) model, the dynamics of capital follow a linear relationship:

$$\hat{k}_{t+1} = g_k \hat{k}_t + g_z \hat{z}_t$$

where g_k and g_z are coefficients that describe the response of capital to its own lag and to TFP shocks, respectively.

The IRF for capital following a one-time innovation $\epsilon_0 = 1$ is:

$$\hat{k}(0) = 0, \quad \hat{k}(1) = g_z \sigma, \quad \hat{k}(2) = g_k \hat{k}(1) + g_z \rho \sigma, \dots$$

This recursive structure simplifies simulations, making it easier to compute the effects of shocks on future variables.

7 Industry Dynamics

7.1 *Hopenhayn model*

The discussion has thus far focused on models with heterogeneous agents; now, attention shifts to models of heterogeneous firms. A prominent example is the model introduced by Hopenhayn (1992), which serves as a foundational framework for understanding industry dynamics. This model captures a steady-state scenario in which firms enter the market, grow, decline, and eventually exit, while the overall distribution of firms remains constant over time. It features an endogenous stationary distribution of firm sizes, making it particularly useful for conducting straightforward comparative statics analyses. Importantly, the model assumes a perfectly competitive market structure, with no strategic interactions among firms.

The framework considers a continuum of firms, each of measure zero, meaning that no individual firm can influence the market price—hence, all firms are price takers. These firms share the same production technology, which exhibits decreasing returns to scale. The model operates under the assumption of no aggregate risk, meaning that both producer prices and factor prices follow a deterministic path and are taken as given by individual firms. Labor is the sole input in the production process. Firms are heterogeneous in their productivity levels, and this productivity is subject to idiosyncratic shocks. These shocks follow a first-order Markov process, which captures the dynamic evolution of firm productivity over time. Entry into the market incurs a fixed cost, and firms also face a constant cost of operation once they have entered.

This modeling framework captures the life cycle of firms within an industry—entry, growth, decline, and exit—while maintaining an equilibrium distribution of firm sizes. Its simplicity and assumptions make it a powerful tool for analyzing how changes in economic conditions, policies, or market structures affect the dynamics of firms and industries.

Incumbent Firm's Problem

The incumbent firm's decision-making process can be described by the following optimization problem:

$$V(z) = \max_n \left(pz f(n) - nw - c_f + \beta \max \left(0, \int v(z') dF(z'|z) \right) \right).$$

Here:

- $f(n)$ is a concave production function, representing output as a function of labor input n .
- $\beta < 1$ is the discount factor, given by $\beta = \frac{1}{1+r}$, where r is the household's discount rate.
- c_f represents the per-period fixed cost of production.
- $F(z'|z)$ is the first-order Markov transition probability function governing productivity shocks.

Timing and Assumptions

The timing of the model assumes that the incumbent firm observes its current state z , takes prices as given, and chooses the labor input n before the realization of the next-period productivity shock z' . Since this is a competitive model, the firm perceives prices p and wages w as exogenous. Consequently, the firm's value function $v(z)$ depends only on the state z , making it invariant to p and w in equilibrium.

The solution to the firm's static optimization problem satisfies the first-order condition:

$$pz f'(n) = w.$$

Given the concavity of $f(n)$, this condition yields a well-defined optimal labor input $n(z)$ for any given productivity level z . Importantly:

- As $f'(n) > 0$, the firm's size (measured by n) is increasing in productivity z .
- The firm's per-period profit, net of fixed costs, is positive:

$$pz f(n(z)) - wn(z) > 0.$$

Exit Rule and Productivity Threshold

Firms exit the market when their productivity falls below a threshold z^* . This threshold z^* is defined such that firms are indifferent between

operating and exiting, i.e., their expected discounted future profits are zero. Specifically:

$$z < z^* \implies \text{Firm exits.}$$

The value of z^* is determined by solving the firm's problem, accounting for the expected discounted future profitability.

Entrant Firm's Problem

The market includes a large number of potential entrants, each of which is ex-ante identical. An entrant faces an entry cost c_e and draws an initial productivity level z from the distribution $G(z)$. The entrant's decision to enter depends on the expected future value of the firm. Formally, the entry condition is:

$$V^e = \int_S V(z) dG(z) \geq c_e,$$

where:

- V^e represents the entrant's expected future profit.
- $G(z)$ is the cumulative distribution function of the initial productivity shock.
- S is the state space of productivity levels.

If $G(z)$ has a lower mean, entrants tend to start small and grow over time, reflecting the dynamic nature of firm growth in response to their initial productivity draw.

Measure of firms

The law of the motion will be of the measure of firm, which will be a function of the transition and measure of entrants.

Definition 5. *Transition function $P_t : S \times \mathcal{B}_S \rightarrow [0, 1]$ is defined as*

$$P_t(s, B) = \int_{z' \in B} F(z'|z) \mathbf{1}(z \geq z^*) dz'$$

where $S = [\underline{z}, \bar{z}]$ is compact subset of individual state space and $\mathcal{B}_S$ is Borel σ -Algebra on S

The law of motion of the measure of firms is given as

$$\mu_{t+1}(B) = \int_S P_t(z, B) d\mu_t(z) + m_{t+1} G(B) \quad \forall B \in \mathcal{B}_S$$

Here m_{t+1} is the mass of the entrant.

Definition 6 (Stationary Equilibrium). *The stationary equilibrium is comprised of Prices p and w ; Value function $v(z)$; Decision rules $n(z)$ and z^* (exit threshold); Measure of operating firms μ ; Mass of entrants m , such that:*

1. **Individual Firm Optimization:** *Incumbent Firms optimize their behavior given prevailing prices.*

Free Entry Condition: *Potential Entrant Firms enter the market if the value of entry v_e equals the entry cost c_e , i.e.,*

$$v_e = \int_{\mathbb{Z}} v(z) dG(z) = c_e$$

where $G(z)$ is the distribution of firm states.

2. **Stationarity and Consistency:** *The stationary measure of firms $\mu(B)$ for any set $B \in \mathbb{B}$ is given by:*

$$\mu(B) = \int_{\mathbb{Z}} SP(z, B) d\mu(z) + mG(B)$$

where $P(z, B)$ is the transition probability from state z to set B , and m is the mass of new entrants.

3. **Market Clearing:**

- *Labor Market: The total labor demand across all firms must equal the labor supply:*

$$\int_{\mathbb{Z}} Sn(z) d\mu(z) = L_S(w)$$

where $n(z)$ is the number of workers hired by a firm in state z , and $L_S(w)$ is the labor supply.

- *Product Market: The total output must match the total demand for goods:*

$$\int_{\mathbb{Z}} Sz f(n(z)) d\mu(z) = Y_D(p)$$

Solution Algorithm

The solution involves the following steps: for simplicity, we assume the product price p is 1.

1. Loop over w : (independent of m)

- Solve the firm's optimization problem and obtain decision rules $n(z; w)$ and exit threshold $z^*(w)$, along with the value function $v(z; w)$.
- Check the free entry condition $v_e(w) = c_e$ and iterate on w until the condition is satisfied.

2. Loop over m : (given equilibrium wage w)

- Compute the stationary measure of firms μ , using the equation:

$$\mu = m(I - P)^{-1}g$$

where μ is a $1 \times N$ vector, P is an $N \times N$ matrix of transition coefficients induced by the Markov chain and the optimal decisions, and g is a $1 \times N$ vector (the discretized version of G)

- Check the labor market condition:

$$L_S(w) = \sum_z n(z; w)\mu(z; m)$$

Iterate on m

7.2 Effects of Higher c_e in the Steady State

In this section, we explore the steady-state implications of an increase in the entry cost, denoted by c_e . This cost represents the barrier to entry that firms must overcome in order to enter the market. An increase in this entry cost can have far-reaching effects on various macroeconomic variables, such as firm dynamics, wages, firm survival thresholds, the age distribution of firms, and the size distribution of firms. The following analysis elaborates on these effects.

1. Impact on the Value of Entry (v_e)

First, a direct consequence of an increase in the entry cost is an increase in the value of entry, v_e . The value of entry corresponds to the present

value of expected profits that a potential entrant can earn by entering the market. When the entry cost rises, firms need to expect a higher return on their investment to justify entering the market. This increase in the required return makes entry less attractive for potential entrants, but those who do decide to enter expect higher returns relative to their costs. Hence, the value of entry increases.

This effect is intuitive because, when barriers to entry are higher, the expected competition is lower, which can lead to higher profit margins for those firms that do manage to enter. This dynamic is consistent with the basic idea that reduced competition (due to fewer firms entering the market) can raise the profitability of individual firms.

2. Lower Wages as a Result of Fewer Firms Entering the Market

The second effect is that wages (w) in the economy are likely to decrease as a result of the higher entry cost. This occurs because a higher entry cost reduces the number of firms entering the market, leading to lower overall competition for labor. As fewer firms enter, the demand for labor is more constrained, and thus wages decrease.

In a competitive labor market, wages are typically determined by the marginal productivity of labor. When fewer firms enter the market, there is less demand for labor, which leads to a decrease in the equilibrium wage rate. The decrease in wages can have broader macroeconomic implications, as lower wages could potentially reduce the incentives for workers to invest in human capital and may also slow down the process of technological adoption by firms.

3. Lower Firm Survival Threshold (z^*) and Negative Selection

An increase in the entry cost has the potential to affect the survival thresholds of firms. In particular, it lowers the exit threshold z^* , which refers to the minimum productivity level that a firm must have to remain in operation. This occurs because, as entry becomes more costly, the composition of firms in the market changes. Specifically, firms that survive longer are typically those that are more productive, and the threshold productivity for exit shifts downward.

The reduction in the firm survival threshold leads to a phenomenon known as "negative selection." In this case, the more productive firms

are less likely to exit the market compared to less productive firms. However, since fewer firms are entering the market due to the higher entry cost, the distribution of firm productivity in the steady state becomes skewed. Lower-wage firms with relatively lower productivity are more likely to exit, leaving only the more productive firms in the market. As a result, the overall efficiency of the firm population improves, but the number of firms operating in the market declines.

4. Firm Aging and Reduction in the Mass of Young Firms

Another key outcome of higher entry costs is the aging of firms. With fewer new firms entering the market, the composition of firms becomes increasingly weighted toward older, established firms. New firms typically have lower productivity at the time of entry, but they contribute to the dynamism of the economy by introducing innovation and competition. When fewer new firms enter, the economy experiences a reduction in the "fresh blood" that keeps the market competitive.

Over time, the firm distribution becomes more heavily skewed towards older firms that have survived for longer periods. This results in a decline in the proportion of young firms in the market, which has important implications for long-run productivity growth. Young firms are often more innovative and capable of adopting new technologies, so their reduced presence in the economy may lead to slower technological progress and less competitive pressure on established firms.

5. Ambiguous Effects on Average Firm Size ($\bar{n}$)

The effect of a higher entry cost on average firm size $\bar{n}$ is not straightforward and can go in opposite directions depending on the prevailing economic conditions and the interaction between firm dynamics and labor market conditions. The effect is, therefore, ambiguous.

- **Larger Firms Due to Lower Wages:** On one hand, higher entry costs tend to lower wages, which may encourage firms to grow larger. When wages decrease, the cost of labor decreases, and firms may find it more profitable to expand their operations. As firms become larger, they can take advantage of economies of scale, reducing their per-unit production costs. Larger firms can also afford to invest more in capital and technology, which can further improve their productivity. In this case, the reduction in

wages due to higher entry costs may encourage firms to become larger, increasing the average firm size in the economy.

- **Smaller Firms Due to Negative Selection:** On the other hand, the higher entry cost may lead to negative selection, where only the more productive firms survive. These firms tend to be more efficient and may be able to grow larger more easily. However, the fewer number of firms that remain in the market may actually result in a smaller average firm size. This is because firms that are more productive may not necessarily choose to grow aggressively, especially if they are already operating at a high level of efficiency. Moreover, since fewer new firms are entering the market, the mass of new firms—which are typically smaller and more dynamic—is reduced, further reducing the average size of firms in the market.

8 Labor Frictions

8.1 *McCall's Model*

8.2 *Mortensen-Pissarides-Diamond Model*

9 Revisiting Growth: Continuous time

Now we will be dealing with the the models with the continuous time. Starting with the Neoclassical Growth Model. Starting with the firm side. Consider $Y(t)$ be a continuous function of time. The inputs are continous in time as well, the technology input, $X(t)$, is non-rival and affects every firm, this may think of as the common productivity parameter. The production function is denoted as $Y(t) = F(K(t), L(t), X(t))$, here F is neo-classical production function, with increasing productivity and decreasing return. Assume Inada conditions are satisfied as well.

To get the sense of the continuous time from the discrete case, we will start by assuming that the $X(t) = X$, later in the stage we will assume $X(t) = e^{xt}X(0)$ that is $\frac{dX(t)/dt}{X(t)} = \frac{X'(t)}{X(t)} = \frac{ze^{xt}X(0)}{e^{xt}X(0)} = x$.

Writing the national income accounting for the open economy with the assumptions that the government spending is exactly negative of net exports, this will give us:

$$Y(t) = I(t) + C(t)$$

Note: $I(t) = K_{t+1} - K_t + \gamma K_t$, if we assume the monthly change, or divide the unit time in 12 parts, we will get the following ($\Delta = 1/12$): $K_{t+\Delta} - K_t = \Delta(I_t - \gamma K_t)$, now consider the limit of $\Delta \rightarrow 0$, we will have,

$$\lim_{\Delta \rightarrow 0} \frac{K_{t+\Delta} - K_t}{t + \Delta - t} = I_t - \gamma K_t$$

The left hand side of the above equation is $\dot{K}(t)$. In the case of the discrete if $I_t = 0$, we will have $K_t = (1 - \gamma)^t K_o = 0$ if $\gamma = 1$. Whereas, in the continuous case if $I_t = 0$, then $\dot{K}(t) = -\gamma K(t)$. Which will provide the following:

$$\begin{aligned} \frac{\dot{K}(t)}{K(t)} = \gamma &\implies \int_0^t \frac{\dot{K}(\tau)}{K(\tau)} d\tau = -\gamma t \\ [\log K(\tau)]_0^t &= -\gamma t \implies K(t) = e^{-\gamma t} K(0) \end{aligned}$$

Full depreciation here is when the $\gamma = \infty$, on the other hand with respect to the discrete case $\gamma \in [0, 1]$.

$$I(t) = \dot{K}(t) + \gamma K(t) \quad \gamma > 0$$

10 Growth

In economic literature, discussions around micro-inefficiencies often focus on issues like poor education or ineffective healthcare systems. However, one might question whether these inefficiencies truly matter in the broader picture. For instance, it's conceivable that while there may be many micro-level failures, the allocation of resources at the macro level could still correct these inefficiencies. Specifically, there are both good and bad firms, and the good firms tend to grow rapidly, whereas the bad firms fade into insignificance. In this context, the inefficiencies within the bad firms may not be critical as long as there are still many firms, including a few successful ones driving overall growth. The goal is to conceptualize this problem and then explore how we can empirically study it. This connects to the core question of growth theory: Why are some countries rich, some poor, and why do some countries grow while others do not? The challenge is to understand how micro-level data can inform our understanding of these broader patterns, bridging theory and empirical analysis.

Aggregate Growth

Neo-classical growth theory suggests that we can explain the overall economic performance of countries by examining an aggregate production function. This function is typically represented as

$$F(A, \bar{K}_P, \bar{K}_H, L)$$

where A stands for productivity, $\bar{K}_P$ is physical capital, $\bar{K}_H$ is human capital, and L is labor. This framework provides a starting point for understanding growth theory.

The question arises: why focus on this aggregate equation? While governance factors do play a role, productivity A is expected to capture some of these effects. The issue is that this model doesn't delve into the disaggregation of economic variables—specifically, it doesn't address who owns the capital and the relationship between capital ownership and individual productivity. Some might argue that technology can shed light on distribution patterns, but the question remains: when can we accurately model this relationship? Essentially, the theory suggests that if two economies have the same levels of these variables, they should have the same GDP, regardless of how capital is distributed.

We are essentially deriving the theorem that ensures the existence of the aggregate production function. The key assumption is that all factor markets are perfect, meaning that individuals can buy or sell any quantity of goods at a given price. In this setup, the social planner allocates a certain amount of inputs based on the productivity θ to maximize the GDP:

$$\max_{\{K_P(\theta), K_H(\theta), L(\theta)\}} \left(\int_{\theta} F(K_P(\theta), K_H(\theta), L(\theta), \theta) d\tilde{G}(\theta) \right)$$

subject to the following resource constraints:

$$\begin{aligned} \int_{\theta} K_P(\theta) d\theta &= \bar{K}_P, \\ \int_{\theta} K_H(\theta) d\theta &= \bar{K}_H, \\ \int_{\theta} L(\theta) d\theta &= \bar{L} \end{aligned}$$

Note that:

1. Under the assumption of perfect markets, the planner's problem is identical to what the market would solve. This reflects the first welfare theorem.
2. It is important to note that the only constraints in this model are the three mentioned above and the distribution of θ . The solution derived will be determined by these constraints, while other factors will not affect it. Specifically, the total quantities in the economy and the distribution of θ are the only aspects that will influence the solution. Therefore, the answer is dependent only on A , $\bar{K}_P$, $\bar{K}_H$, and $\bar{L}$.

Theorem: The sufficient condition for the existence of the aggregate production function is that all markets are perfect.

This condition is sufficient, but not necessary. Why is this the case? Consider the following example: if we take all capital and allocate it equally to everyone, as well as all other inputs, this would still generate output. However, this approach would also yield an aggregate production function, even though it is a very inefficient one, because both the productive and non-productive individuals would receive the same capital. Conversely, any assignment rule that does not generate

perfect markets would not result in an aggregate production function.

The distribution of wealth does not affect the construction of the production function. Why is this? It is because perfect markets imply that who owns the capital does not matter for its use. In other words, the ownership of capital does not influence its productivity in the model.

The relationship between individual and aggregate production functions is often questioned, particularly in terms of properties like diminishing returns to scale and convexity. The short answer is NO, there is no direct, simple relationship.

Theorem: Convexification by Aggregation (Shapley-Folkman-Starr) Let $S_1, S_2, \dots, S_n$ be subsets of a Euclidean space $\mathbb{R}^k$, and let the sets $S_1, S_2, \dots, S_n$ be convex for each $i = 1, 2, \dots, n$. Consider the Minkowski sum of these sets, defined as:

$$S = \sum_{i=1}^n S_i = \left\{ \sum_{i=1}^n x_i \mid x_i \in S_i \right\}.$$

The Shapley-Folkman-Starr theorem states that if the number n of sets is large enough, the Minkowski sum S will be close to a convex set. More precisely, for a large n , the Minkowski sum of the convex sets $S_1, S_2, \dots, S_n$ is ****almost convex**** in the sense that the set of points in the Minkowski sum will approximate a convex set in the following way.

To understand this, consider an individual production function that follows an S-shaped curve. In this case, the function is increasing at first but becomes concave only after a certain level of input. This suggests that, while the function is not concave everywhere, it becomes concave beyond a particular point. Despite this, it can still be argued that the aggregate production function will exhibit concavity. The key insight here is about efficiency in resource allocation. The goal is to assign capital in such a way that the average product is maximized for each firm, which, in turn, maximizes the overall production potential. In other words, if there are two firms, to maximize their total output, the capital should be distributed between them in a manner that maximizes the sum of their average products. In the case of the S-shaped individual production function, the point where the average product is maximized is tangent to the production function at a point where

it has already become concave. This implies that even though the individual production function may not be concave across all levels of input, the aggregate production function, derived from optimizing the allocation of capital across firms, will exhibit concavity. Thus, while individual production functions may not necessarily follow a concave shape in every region, the aggregate production function, built upon efficiency considerations, is likely to be concave due to the way resources (capital) are allocated to maximize output.

11 Appendix - A

In this appendix we will introduce and discuss some basic concepts from Probability and optimization.

11.1 Basic concepts in Stochastic Process

Definition 7. A probability space is a mathematical object consisting of three elements:

1. A set Ω of all possible outcomes ω
2. A collection $\mathcal{F}$ of subsets of Ω that constitute the events to which probability is assigned (a σ -algebra)
3. A set valued mapping $\mathbf{P}$ that assigns probability values to those events (elements of $\mathcal{F}$).

A Probability space is denoted by

$$(\Omega, \mathcal{F}, \mathbf{P})$$

Definition 8. A σ -algebra $\mathcal{F}$ is a special kind of family of subsets of a space Ω that satisfy three properties:

1. $\Omega \in \mathcal{F}$,
2. $\mathcal{F}$ is closed under complementation: $E \in \mathcal{F} \rightarrow E^c \in \mathcal{F}$,
3. $\mathcal{F}$ is closed under countable union: if $\{E_i\}_{i=0}^{\infty}$ is a sequence of sets such that $E_i \in \mathcal{F} \forall i$, then $(\cup_{i=0}^{\infty} E_i) \in \mathcal{F}$,

Definition 9. A random variable is a function whose domain is the set of events Ω and whose image is the real numbers (or a subset of real numbers):

$$x : \Omega \rightarrow \mathbf{R}$$

For any real number α define the set (E_α) of all elements of the sample space such that the function x takes the value less than α :

$$E_\alpha = \{\omega : x(\omega) < \alpha\}$$

Let us now define when does a function said to be measurable. The concept of measurability is needed so that we can assign probabilities to the events like $x < \alpha$ for all real number α

Definition 10. A function x is said to be measurable with respect to the σ -algebra $\mathcal{F}$ (or sometimes denoted as $\mathcal{F}$ -measurable) if the following holds:

$$\forall \alpha \in \mathbf{R} : E_\alpha \in \mathcal{F}$$

Definition 11 (Existence of Stationary Distribution). Let $X = (X_t)_{t \geq 0}$ be a discrete-time Markov chain with state space $\mathcal{S}$ (finite or countable), and let $P = (p_{ij})$ be the transition probability matrix, where $p_{ij} = \mathbb{P}(X_{t+1} = j \mid X_t = i)$ is the probability of transitioning from state i to state j . Assume the following conditions:

1. **Irreducibility:** The chain is irreducible, meaning that for any two states i and j , there exists a finite number of steps such that the chain can transition from i to j .
2. **Aperiodicity:** The chain is aperiodic, meaning that there is no integer $d > 1$ such that every state can only return to itself in multiples of d steps.
3. **Positive Recurrence:** The chain is positive recurrent, meaning that the expected return time to any state is finite.

Then, there exists a unique stationary distribution $\pi = (\pi_1, \pi_2, \dots, \pi_n)$ such that:

$$\pi P = \pi$$

and

$$\sum_{i \in \mathcal{S}} \pi_i = 1, \quad \pi_i \geq 0 \quad \forall i \in \mathcal{S}.$$

This distribution represents the long-run distribution of states in the Markov chain.

12 Appendix - B

12.1 Introduction

When agents/firms are uncertain about the future, solving their maximization problems often requires computing a conditional expectation. For instance, consider a firm optimization problem where a firm faces stochastic productivity shocks that follow an AR(1) process. The firm's goal is to choose the optimal level of capital k' for the next period, given its current capital k , current productivity shock z , and other parameters.

The firm's value function is given by:

$$V(k, z) = \max_{k'} \{ \pi(k, z) - k' + \beta E[V(k', z')|z] \},$$

where: - $\pi(k, z) = zf(k) - wk$ is the firm's profit function, with $f(k)$ representing the production function, - β is the discount factor, - k' is the capital chosen for the next period, - z is the firm's productivity, which evolves according to the AR(1) process:

$$z' = (1 - \rho)\mu_z + \rho z + \epsilon, \quad \epsilon \sim N(0, \sigma_\epsilon^2),$$

and w is the wage rate.

Discretizing the State Space

To impute the integral of the value function weighted by the density $f(z'|z)$, i.e.,

$$E[V(k', z')|z] = \int_{-\infty}^{\infty} V(k', z') f(z'|z) dz'$$

The main idea to impute this integral is to discretize the state space, here the productivity shock z is discretized into a finite set $Z = \{z_1, z_2, \dots, z_n\}$. The transition probabilities between different productivity levels are captured by the transition matrix P , where $P_{i,j}$ represents the probability of moving from z_i to z_j .

The conditional expectation $E[V(k', z')|z]$ can then be computed as:

$$E[V(k', z')|z_i] = \sum_{j=1}^n P_{i,j} V(k', z_j).$$

Now, the question is to what is the ways to discretize the state space, which are efficient one?

12.2 Discretizing AR(1)

12.2.1 Tauchen

In 1986, Tauchen [?] introduced a method to approximate processes based on the assumption that, conditional on a given value of z , the next-period value z_0 follows a normal distribution. The conditional mean of z_0 is given by $(1 - \rho)\mu_z + \rho z$, and the standard deviation is σ_ϵ , which represents the uncertainty in this process.

The method involves discretizing the continuous state space by choosing a set of values, z , which are evenly spaced along the real line. These values, denoted as $z_1, z_2, \dots, z_n$, represent the possible states that z can take. The first value, z_1 , is the smallest, and z_n is the largest. The selection of these bounds, z_1 and z_n , is important, and they are set at m unconditional standard deviations away from the unconditional mean of z .

This means:

$$z_1 = \mu_z - m\sigma_z$$

and

$$z_n = \mu_z + m\sigma_z$$

where the unconditional standard deviation σ_z is defined as:

$$\sigma_z = \frac{\sigma_\epsilon}{\sqrt{1 - \rho^2}}$$

Next, the probability transition matrix, which determines the likelihood of transitioning from one state to another, is constructed. The difference between consecutive values of z , denoted by w , is:

$$w = z_j - z_{j-1}$$

For j between 2 and $n - 1$, the transition probability from z_i to z_j , denoted as $p_{i,j}$, is computed as:

$$p_{i,j} = \text{Prob} \left[z_j - \frac{w}{2} \leq (1 - \rho)\mu_z + \rho z + \epsilon \leq z_j + \frac{w}{2} \right]$$

For the first state ($j = 1$), the transition probability is:

$$p_{i,1} = \text{Prob} \left[(1 - \rho)\mu_z + \rho_z + \epsilon \leq z_1 + \frac{w}{2} \right]$$

Finally, for the last state ($j = n$), the transition probability is:

$$p_{i,n} = 1 - \text{Prob} \left[z_n - \frac{w}{2} \leq (1 - \rho)\mu_z + i + \epsilon \right]$$

This setup provides a discrete approximation to the continuous process, allowing for the computation of transition probabilities across the chosen grid of states. The accuracy of this approximation improves as the number of states, n , increases. But this method does not work well for the highly persistent process.

12.3 Rouwenhorst

For highly persistent processes, where $\rho > 0.9$, it is often beneficial to use Rouwenhorst's discretization method [?]. This method is especially useful in approximating processes with high persistence, preserving the first-order serial correlation and the variance of the continuous process in the discrete state space.

Let z consist of n points that are symmetrically and evenly spaced over the interval $[\mu_z - \nu, \mu_z + \nu]$, where ν controls the spread of the points, capturing the variability of z .

The probability transition matrix P is then constructed as follows:

1. Choose parameters p and q .
2. When $n = 2$, the probability transition matrix is:

$$P_2 = \begin{bmatrix} p & 1 - p \\ 1 - q & q \end{bmatrix}$$

3. For $n = 3$, the transition matrix P_3 becomes:

$$P_3 = \begin{bmatrix} p^2 & 2p(1 - p) & (1 - p)^2 \\ p(1 - q) & pq + (1 - p)(1 - q) & q(1 - p) \\ (1 - q)^2 & 2q(1 - q) & q^2 \end{bmatrix}$$

4. For larger n , the transition matrix P_n is derived recursively from P_{n-1} using the following rule. First, compute the sum of the following four matrices:

$$p \begin{bmatrix} P_{n-1} & 0 \\ 0 & 0 \end{bmatrix} + (1-p) \begin{bmatrix} 0 & P_{n-1} \\ 0 & 0 \end{bmatrix}$$

$$(1-q) \begin{bmatrix} 0 & 0 \\ P_{n-1} & 0 \end{bmatrix} + q \begin{bmatrix} 0 & 0 \\ 0 & P_{n-1} \end{bmatrix}$$

Here, 0 is a $(n-1) \times 1$ column vector, and $0'$ is a $(1 \times n-1)$ row vector. Once the matrices are added, all but the first and last rows are divided by 2 to ensure that the rows sum to 1, maintaining the correct conditional probabilities.

Interpreting the Transition Probabilities

The parameters p and q play key roles in determining the transition probabilities:

- p^{n-1} is the probability of remaining in the lowest state once you are already there.
- q^{n-1} is the probability of remaining in the highest state.
- Conversely, $(1-p)^{n-1}$ represents the probability of transitioning from the lowest to the highest state, while $(1-q)^{n-1}$ is the probability of transitioning from the highest to the lowest state.

Choosing $p \neq q$ introduces conditional heteroscedasticity in the shocks. It can be shown that regardless of the specific values of n and z , the first-order serial correlation of this process is given by:

$$\text{First-order serial correlation} = p + q - 1$$

Thus, if $p = q = \pi$, then the value of π can be chosen so that the discrete process has the same first-order persistence as the continuous process.

Variance of the Process

Additionally, the variance of the discretized process $\tilde{z}$ is:

$$\text{Var}(\tilde{z}) = \frac{\nu^2}{n-1}$$

Therefore, for a given n , a good choice of ν is one that ensures the variance of $\tilde{z}$ matches the variance of the continuous process z . This leads to the following expression for ν :

$$\nu = \sqrt{\frac{n-1}{\rho^2-1}}\sigma_\epsilon$$

where σ_ϵ is the standard deviation of the shocks.

13 *A Few Problems with proposed solutions:*

Q1) Setup and solve for the planner's problem in 2 agent economy with CRRA preferences under the complete market with Arrow-Debreu time-0 markets. Comment on the deterministic case as well.

Consider a consumer with a CRRA utility function and maximize the expected life-time utility over the finite state space.

$$E \left[\sum_{t=0}^{\infty} \beta^t \frac{c_{it}(s^t)^{1-v} - 1}{1-v} \right] = \sum_{t=0}^{\infty} \sum_{s^t} \beta^t \frac{c_{it}(s^t)^{1-v} - 1}{1-v} \pi(s^t)$$

Here, s^t is the state (including the history of states up to time t), $c_{it}(s^t)$ is the consumption of agent i at time t , $\pi(s^t)$ is the probability of the realization of state s^t from the viewpoint of time 0. We now assume that the markets are complete, that is at time 0, the market for contingency claims or Arrow securities for all states s^t opens. let, $p_t(s^t)$ be the time-0 price of an Arrow security that promises to pay one unit of consumption good at time t if the state s^t realizes. The budget constraint of the agent is,

$$\sum_{t=0}^{\infty} \sum_{s^t} p_t(s^t) c_{it}(s^t) \leq \sum_{t=0}^{\infty} \sum_{s^t} p_t(s^t) w_{it}(s^t) = W_{i0}(\mathbf{p})$$

Here, $W_{i0}(\mathbf{p})$ is the wealth of the agent i at time 0, $w_{it}(s^t)$ is the state dependent endowment of agent i at time t if the state is s^t .⁹ The Euler equation is:

$$\frac{1}{p_t(s^t)} \pi(s^t) c_{it}(s^t)^{-v} = \beta \frac{1}{p_{t+1}(s^{t+1})} \pi(s^{t+1}) c_{i,t+1}(s^{t+1})^{-v}$$

It is to notice that c_{it} can be rewritten as a function of c_{i0} . Then c_{i0} can be solved using the budget constraint. Consumption of agent i at each period can be expressed as $c_{it}(s^t) = g(s^t, \mathbf{p}) W_{i0}(\mathbf{p})$. Thus the consumption for consumers i and j satisfies the equation given the function g is not dependent on i . Therefore, taking the ratio of the

⁹What happens in deterministic case is the following:

The budget constraint becomes, $a_{i,t+1} + c_{it} \leq y_{it} + (1+r)a_{it}$, and given a_0 and $\lim_{T \rightarrow \infty} a_T / (1+r)^T = 0$ and can be written as, $\sum_{t=0}^{\infty} \frac{1}{(1+r)^t} c_{it} \leq \sum_{t=0}^{\infty} \frac{1}{(1+r)^t} y_{it}$

consumption for consumers i and j , we get

$$\frac{c_{it}(s^t)}{c_{jt}(s^t)} = \frac{W_{i0}(\mathbf{p})}{W_{j0}(\mathbf{p})}$$

Plugging this into the aggregate constraint, we get,

$$C_t(s^t) = \sum_i c_{it}(s^t) = g(s^t, \mathbf{p}) \sum_i W_{i0}(\mathbf{p}) = p_t(s^t) g(s^t, \mathbf{p}) \sum_i w_{it}(s^t)$$

The above implies that the distribution does not matter in the aggregation or for the macroeconomic outcome. Note, using the market clearing condition for the equilibrium, the prices will be determined by equating $p_t(s^t) g(s^t, \mathbf{p}) = 1$ for each state.

The ratio of the consumption of individual i and j tells us that the policy change affects the rich and the poor in the same direction, and unless the policy is different for rich and poor. This is one of the motivation to study the incomplete markets to pin down the wealth distribution to a stationary wealth distribution and is unique.¹⁰ In the complete market scenario, we can not model or talk about the insurance policy on idiosyncratic shocks/risks. For example, the unemployment insurance, government-sponsored health insurance, etc. In real scenarios agents are not fully insured, and can access only to self-insurance, through some type of asset, bonds.

¹⁰The existence of the unique stationary distribution is in Appendix ??

Some Practice Problems

Proposed solutions

Solution 1

(a) Production and preferences

$$y = zk^\alpha n^{1-\alpha}, \alpha \in (0, 1)$$

$$u(c) = \begin{cases} c^{1-\gamma}/1-\gamma & \text{if } \gamma \neq 1 \\ \log c & \text{if } \gamma = 1 \end{cases}$$

Problem in recursive form: Consumer solves:

$$V(k, K) = \max_{c, k'} \{U(c) + \beta V(k', K')\}$$

assuming the following:-

(i) utility is concave, agent is infinitely lived and discount the next period by $\beta \in (0, 1)$. (ii) production function is neoclassical production function, firms own nothing, therefore faces the static problem given the labor supply and capital supply by household at a given point in time, The firm problem is

$$\max_{k_t^d, n_t^d} \{F(k_t^d, n_t^d) - r_t k_t^d - \omega_t n_t^d\} \forall t$$

(iii) k_0 is given, Household make decision (dynamically) of saving and consumption.

(iv) n does not enter agent's/consumer's utility. Therefore $n = 1$ and hence, when firm solves static problem.

$$\Rightarrow F_k(K_t, 1) = R(K)$$

from $f'(k_t) = 1 - \delta + r_t(k)$ [$f(k_t) = (1 - \delta)k_t + F(k_t)$]

(v) Budget constraints. $c + k' = [r(K) + 1 - \delta]k$.

$$c, k' \geq 0$$

Given the assumptions, our constraint set is compact, Objective function is continuous, concave as well. The solution to the consumer problem will be unique.

(b) Here our state space is compact subset of $\mathbb{R}$ and therefore, first thing to do for numerical computing the problem is to discretize the set and then iterating the value function on it. Formally:- (i)

transforming $X = [0, \max \{k_0, \bar{k}\}]$ to $K = \{k_1, k_2, \dots, k_n\}$ with k_i as the extreme points of the intervals.

$$(k_1 < k_2 < \dots < k_n)$$

(ii) Set $k_1 = k_0$ (Given) and $k_n = k^*$, choose n such that increasing the intervals $(n + 1)$ will keep the value in ϵ nod. (iii) Applying recursive method of seccusive approximate ⁿ

$$V_{j+1}(k) = \max_{k' \in [0, f(k) + (1-\delta)k], k' \in K} [u(f(k) + (1-\delta)k - k') + \beta j'(k)]$$

where V_0 is initial guess (arbitrary) iterate un till the $V_{j+1}(k) - V \cdot (k) < \varepsilon$

We know this value function will converge, as Bellman equation is a contraction mapping and using contraction mapping theorem in the Banach space (knicuing bovid payoff fractions), we know there exists a unique fixed point

(c) Solving for the general case:

$$\begin{aligned} V(k', k) &= \max_{C, k'} \{u(c) + \beta V(k', k')\} \\ FOC &\Rightarrow U'(C(k, k)) = \beta [R(k') + 1 - \delta] u'(C(k', k')) \\ C(k) + k' &= (R(k) + 1 - \delta)k + \omega(k) \\ R(k) &= F_k(k, 1) \\ W(k) &= F_w(k, 1) \end{aligned}$$

representative agent $k = K$ (along the eq path)

$$\begin{aligned} &\Rightarrow u'(C(k)) = \beta [R(k') + 1 - \delta] u'(c(k')) \\ C(k) + k' &= (R(k) + 1 - \delta)k + \omega(k) \\ R(k) &= F_k(R, 1) = f'(k) = z\alpha k^{\alpha-1} n^{1-\alpha} \\ W(k) &= F_N(k, 1) = z(1 - \alpha)k^\alpha n^{-\alpha} \\ &\Rightarrow C(k) + k' = [z\alpha k^{\alpha-1} n^{1-\alpha} + 1 - \delta] k + z(1 - \alpha)k^\alpha n^{-\alpha} \end{aligned}$$

also

$$u'(C(k)) = \beta [z\alpha k'^{\alpha-1} n'^{1-\alpha} + 1 - \delta] u'(C(k'))$$

for $u = \log, \delta = 1, z = 1$

$$\begin{aligned} C(k) &= [\alpha k^{\alpha-1} n^{1-\alpha}] k + (1 - \alpha) k^\alpha n^{-\alpha} - k' \\ &= \alpha k^\alpha n^{-\alpha} + (1 - \alpha) k^\alpha n^{-\alpha} - k' \\ &= k^\alpha n^{-\alpha} [n\alpha + 1 - \alpha] - k' \end{aligned}$$

Euler is now,

$$k'^\alpha (n')^{-\alpha} [1 + (n' - 1)\alpha] - k'' = \beta [\alpha (k')^{\alpha-1} (n')^{1-\alpha}] [k^\alpha n^{-\alpha} (n\alpha + 1 - \alpha) - k']$$

The above is the second order difference equation and can be solved using the intital and transversality conditions.

In the Steady state $k' = k'' = k$ and $n' = n$

$$\begin{aligned} k^\alpha n^{-\alpha} [1 + (n - 1)\alpha] - k &= \beta (\alpha k^{\alpha-1} n^{1-\alpha}) [k^\alpha n^{-\alpha} \cdot (n\alpha + 1 - \alpha) - k] \\ \Rightarrow \left(\frac{k}{n}\right)^{1-\alpha} &= \alpha\beta \\ k &= (\alpha\beta)^{\frac{1}{1-\alpha}} n^{1-\alpha} \end{aligned}$$

with, z

$$k = (z\alpha\beta)^{\frac{1}{1-\alpha}} n^{1-\alpha}$$

The value function iteration is shown in the graph below (Figure 1), with certain value function iterations. It can be seen that the value function is converging. Now the below shows the policy function for the case when $\delta = 1$ and $\gamma = 1$

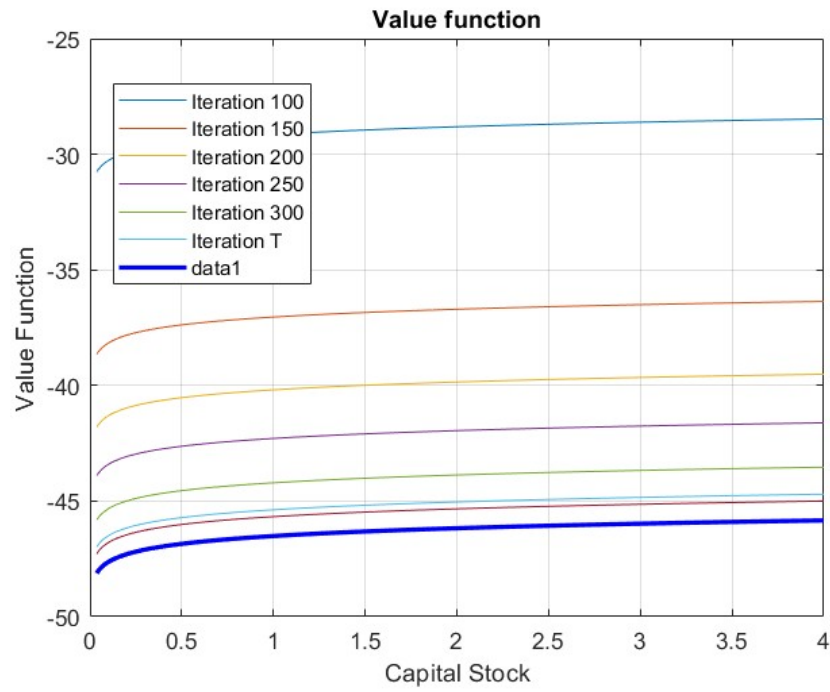


Figure 1:

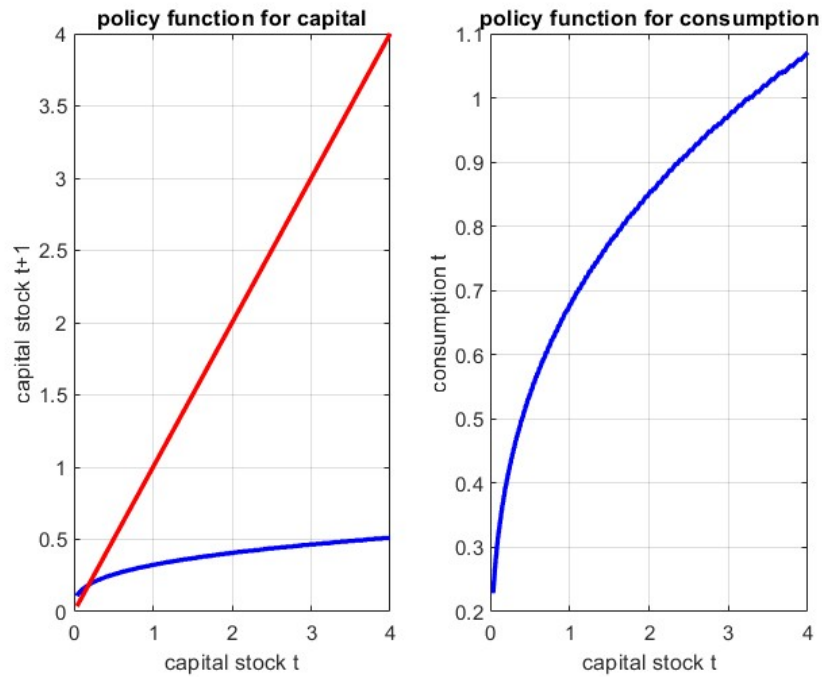


Figure 2:

- (d) The graph shows the case for the original parameters. When $\delta = 0.1$ and $\gamma = 1.5$. Value function iteration are shown in the

Figure 3, and policy function is shown in the Figure 4.

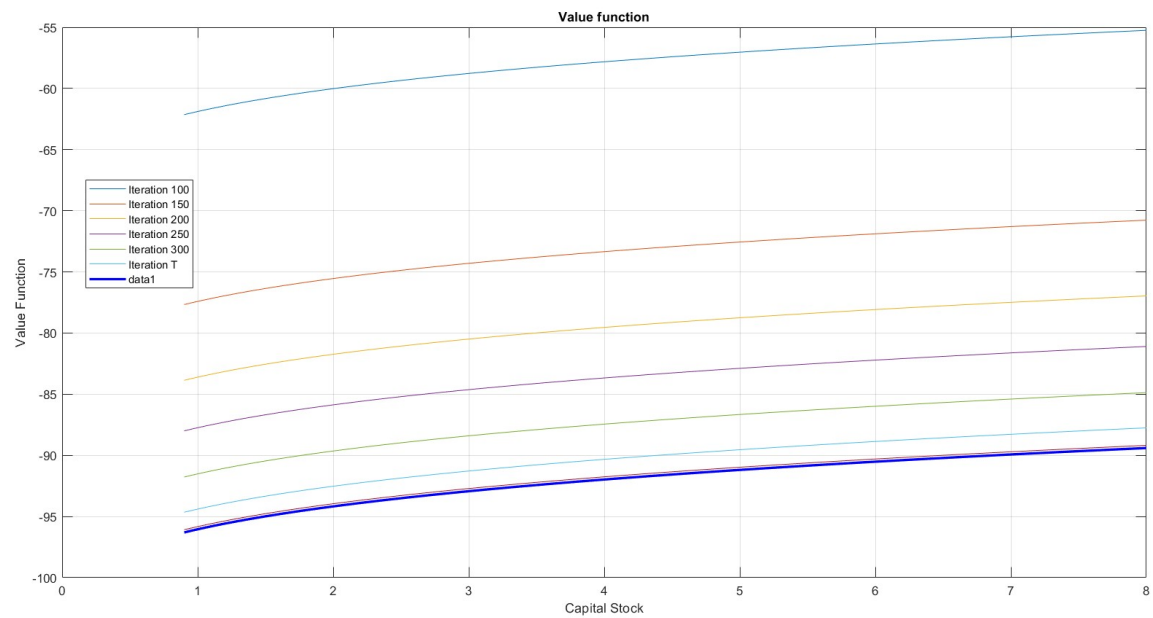


Figure 3:

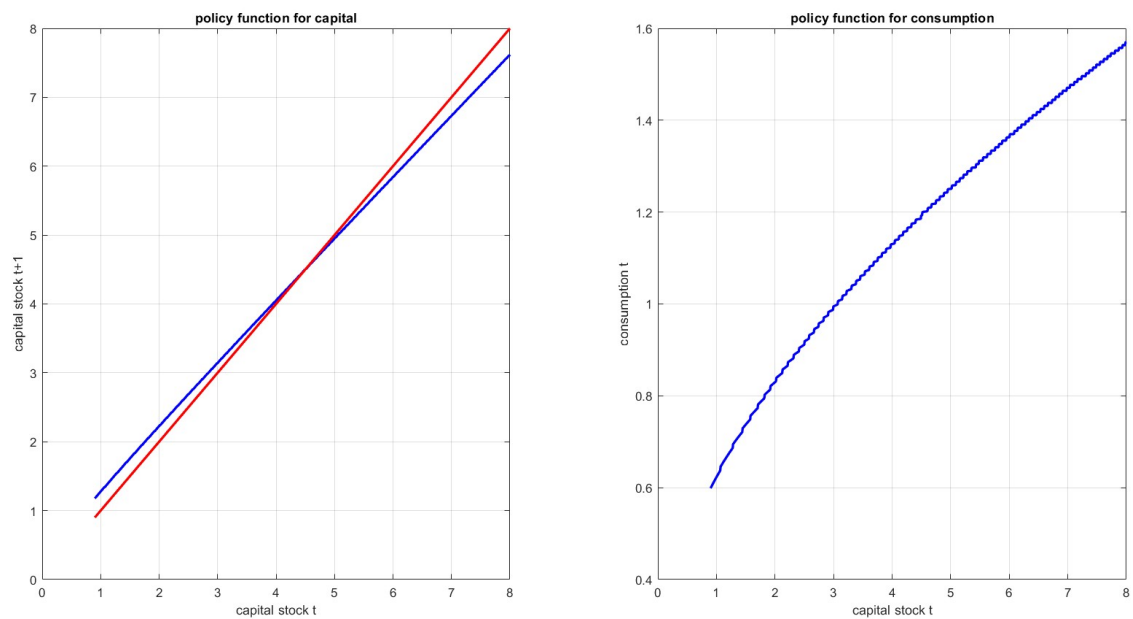


Figure 4:

(e) Now the graph below shows that transition dynamics of the capital, consumption, and output. It can be seen that the transition

converges to the steady state and is a decreasing sequence with respect to the time period.

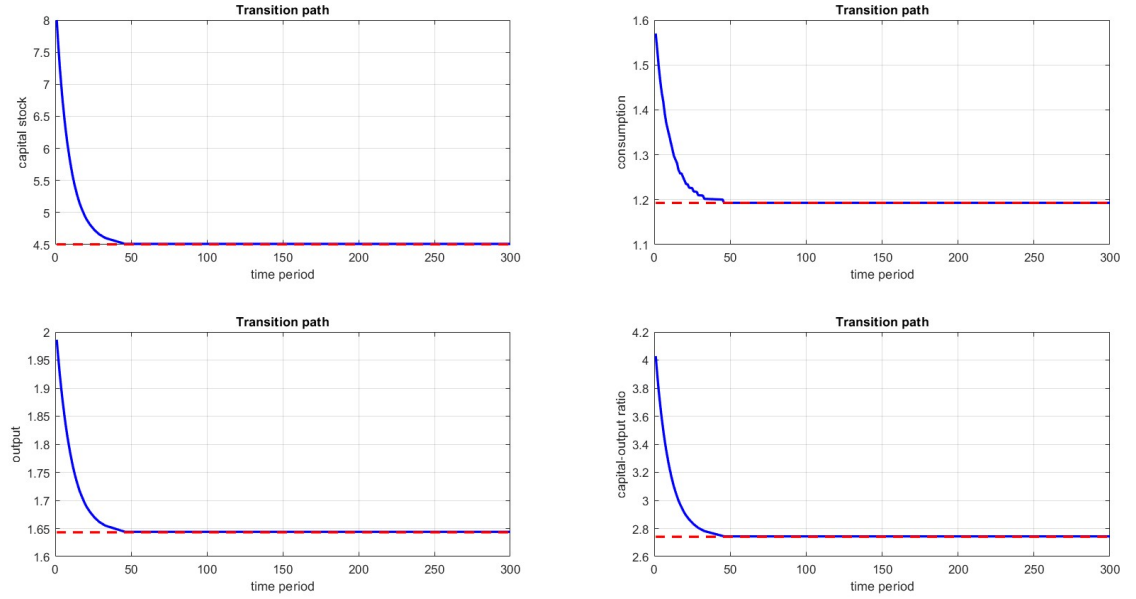


Figure 5:

(f)

(g) If $\alpha \uparrow \implies (\alpha\beta) \uparrow \implies \left(\frac{1}{1-\alpha}\right) \uparrow \implies k^* \uparrow$, this implies that if the share of the capital in the production increases the steady state capital will also increase as increase in the share will lead to marginal increase in the output and the steady state capital will be higher.

if $z \uparrow \implies k^* \uparrow$, similar to the case when capital share increases, when the technology becomes more productive, the steady state capital increases, this can be seen from the steady state capital as a function of z .

if $\beta \uparrow \implies k^* \uparrow$, If β increases it implies that the the agent worries more about the future and now saves more. This implies that in the steady state it will save more as well and k^* will be higher.

(h) Value Function Iteration:

- 1) Initializing the value function array.
- 2) Performing the value function iteration: For each grid point,

calculate the expected future value using piece wise linear interpolation.

3) Update the value function based on the Bellman equation.

4) Repeat until convergence.

For the case of the Policy Function: 1) Determine the optimal policy function for capital investment by maximizing the expected utility. 2) The policy function represents the optimal investment decision at each capital level.

Yes, this method helped in achieving a good approximation to the true solution with a lower number of grid points. It is mainly possible by employing more sophisticated interpolation techniques or adaptive grid methods.

One that I can think of is to use adaptive grid methods, where the grid points are denser in regions where the function is changing rapidly and sparser where it is changing slowly. This helps in capturing the curvature. This allows for a more efficient allocation of computational resources, focusing on areas of interest.

Solution 2

- (a) The individual consumption is dependent on the history but it does not matter as, the economy is endowment economy with complete market and agents can fully insure themselves. The distinction matters when the markets are incomplete, that is when the number of assets available are less than the number of states.
- (b) Let the weights are $\{\lambda_i\}_{i=1}^I$. Given these weights, the planner solves the following problem:

$$\sum_{i=1}^I \lambda_i \sum_{t=0}^{\infty} \sum_{s^t \in S^t} \beta^t \pi(s^t) u(c^i(s^t))$$

s.t.

$$\sum_{i=1}^I c_i(s^t) = \sum_{i=1}^I y_i(s^t) \quad \forall t, s^t$$

Under the symmetric weights the planner will allocate the equal proportion of the aggregate income to each of the individuals, this

implies that

$$c_i(s^t) = \frac{\sum_{i=1}^I y_i(s^t)}{I} = \frac{Y(s^t)}{I} \quad \forall i \quad \forall t$$

- (c) The ratio of the first order condition or the Euler equation of the planner's problem (after setting up the Lagrangian) for agent i and agent j is:

$$\frac{u'(c^i(s^t))}{u'(c^j(s^t))} = \frac{\lambda_j}{\lambda_i}$$

The left hand side shows the need of the individuals and the right hand side shows the merits. Needs is basically how much both want and merits is based on weights and determines how much they deserve.

$$\implies u'(c^i(s^t)) = \frac{\lambda_j}{\lambda_i} u'(c^j(s^t)) \quad \forall i \neq j$$

Assuming utility functions are continuously differentiable, we have,

$$c^i(s^t) = (u')^{-1} \left[\frac{\lambda_j}{\lambda_i} u'(c^j(s^t)) \right] \quad \forall i \neq j$$

Plugging in the constraint of the planner problem, we have

$$\sum_{i=1}^I c^i(s^t) = \sum_{i=1}^I y^i(s^t)$$

$$\sum_j (u')^{-1} \left[\frac{\lambda_j}{\lambda_i} u'(c^j(s^t)) \right] = Y(s^t) \quad \forall j$$

The last equality implies that the consumption of j is only a function of $Y(s^t)$ and is independent of the j 's income. This shows that under socially optimally, every agent is fully insured, or there is a perfect risk sharing.

- (d) Since, there is a technology to save, the planner's problem is now as following:

$$\sum_{i=1}^I \lambda_i \sum_{t=0}^{\infty} \sum_{s^t \in S^t} \beta^t \pi(s^t) u(c^i(s^t))$$

s.t.

$$\sum_{i=1}^I c_i(s^t) - a_{t+1}(s^t) = \sum_{i=1}^I y_i(s^t) + (1 + r(s^t))a_t \quad \forall t, s^t$$

Note that a_0 is zero.

- (e) Solving the planner's problem now will give two sets of First order conditions, one with respect to the consumption at time t for state s^t and second is with respect to the a_{t+1} . These are:

$$\lambda_i \beta^t \pi(s^t) u'(c^i(s^t)) = \theta(s^t) \quad \forall t \quad \forall s^t \quad \forall i$$

and

$$\theta(s^t) = \theta(s^{t+1})[1 + r(s^{t+1})] \quad \forall t \quad \forall s^t$$

Here $\theta(s^t)$ is the Lagrangian multiplier.

The Lagrangian multiplier for individual i and individual j is same, which gives us.

$$\frac{u'(c^i(s^t))}{u'(c^j(s^t))} = \frac{\lambda_j}{\lambda_i} \quad (7)$$

Note that this is same as before with no saving technology. Though here the resource constraint is different. Combining the two FOC's gives us:

$$u'(c^i(s^t)) = \beta[1 + r(s^{t+1})] \frac{\pi(s^{t+1})}{\pi(s^t)} u'(c^i(s^{t+1})) \quad (8)$$

- i) Assuming that the aggregate income is constant ($\bar{Y}$) and $1 + r(s^{t+1}) = \beta^{-1}$. The equation (7) remains unchanged and (8) becomes,

$$u'(c^i(s^t)) = u'(c^i(s^{t+1})) \quad (9)$$

$$\implies c^i(s^t) = \frac{\lambda_i}{\sum_i \lambda_i} \bar{Y} \quad (10)$$

This implies that $a_{t+1} = 0$, for all t . This is intuitive in the sense that if the income is fixed overtime over all states, the consumption is smoothed for each agent by allocating them the proportion (=weight) of the aggregate income at each period and for each state.

- ii) A deterministic time trend in the aggregate endowment. If the aggregate income endowment becomes deterministic, t comes in the picture. Logically, the planner would like to same at the time of high aggregate endowment for the state with low aggregate en-

dowment. This also implies that if the aggregate endowment is monotonically increasing with time, then the planner would not like to save, and $a_t = 0 \forall t$.

In this case the consumption will still be the function of aggregate endowment, since the a_{t+1} might not be zero, based on sequence Y_t the consumption of individuals becomes the function of a_{t+1} as well.

$$c^i(s^t) = (u')^{-1} \left[u'(c^j(s^t)) \frac{\lambda_j}{\lambda_i} \right] \quad (11)$$

Using the resource constraint (summing over all individuals) in the above equation we have,

$$\sum_{i=1}^I c^i(s^t) = \sum_{i=1}^I (u')^{-1} \left[u'(c^j(s^t)) \frac{\lambda_j}{\lambda_i} \right] = \sum_{i=1}^I y_i(s^t) + (1 + r(s^t))a_t - a_{t+1}(s^t) \quad (12)$$

$$\implies c^j(s^t) = c^j(Y_t, a_t, a_{t+1}) \quad (13)$$

Note that the last expression implies that the consumption of individual j is a function of aggregate endowment (time dependent) and saving a_{t+1} and return a_t .

iii) Similarly for the case when income is dependent on the state, the value of a_{t+1} will not be zero, as the planner would like to save for bad times. Since there is no production in the economy, the uncertainty of the aggregate endowment comes from the individuals' state-dependent endowment. The Planner wants to smooth out the consumption and the planner would make the decision of saving (a_{t+1}) at time t which will be deducted from the aggregate endowment of the income and also the return from the last period savings. In this case, consumption will be a function of the aggregate endowment and the savings and returns.

Same as the expression in the equation (12). The consumption of agent j will be function of aggregate endowment (state dependent).

$$c^j(s^t) = c^j(Y(s^t), a_t(s^t), a_{t+1}(s^t), r(s^t)) \quad (14)$$

- (f) It can be seen from above that the individual consumption is a function of aggregate income (endowment + return from investments). This implies that the individual consumption is history independent (separable) and independent from the cross-sectional endowment distribution.
- (g) Considering $u(c) = \frac{c^{1-\sigma}}{1-\sigma}$. As solved in the above parts, we have,

$$\begin{aligned}
c^i(s^t) &= (u')^{-1} \left[u'(c^j(s^t)) \frac{\lambda_j}{\lambda_i} \right] \\
&= \frac{\lambda_j^{1/\sigma} c^j(s^t)}{\lambda_i^{1/\sigma}} \\
\sum_j c^j(s^t) &= \frac{c^i(s^t)}{\lambda_i^{1/\sigma}} \sum_j \lambda_j^{1/\sigma} \\
c^i(s^t) &= \frac{\lambda_i^{1/\sigma}}{\sum_j \lambda_j^{1/\sigma}} C(s^t) \\
&= \alpha^i C(s^t)
\end{aligned}$$

Solution 3

- (a) Complete Financial market: Given K assets and S states of the world. Let, $\mathbf{a}_{t+1}$ be the asset portfolio of an individual at the end of the time t with some gross pay-off for each state of the world (a matrix of $S \times K$). When consumer can generate any financial income for tomorrow based on the appropriate choice of portfolio is known as Complete financial markets.

Here, we have 2 assets and 2 states of the world and therefore the financial markets are complete. Though the markets are incomplete as there are aggregate shocks and that are not insurable. If we add one more state of the nature, then the financial markets will be incomplete and market will incomplete as well.

$$R = \begin{array}{c|cc} & \text{Insuarance} & \text{Share} \\ \hline \text{L} & 1 & y_l \\ \hline \text{H} & 0 & y_h \\ \hline \end{array}$$

(b) The problem of the representative consumer is :

$$\begin{aligned} & \max_{c_t} \mathbf{E}_0 \sum_{t=0}^{\infty} \beta^t \ln(c_t) \\ & s.t. \\ & c_t + p_t s_{t+1} + q_t b_{t+1} = b_t \cdot \mathbf{1}(y_t = y_l) + s_t(y_t + p_t) \end{aligned}$$

(c) A competitive equilibrium with sequential markets is a set $\{p(z^t), q(z^t)\}_{t=0, z^t \in Z^{t+1}}^{\infty}$ of prices and a set of allocation $\{s_i(s^t), c_i(s^t), b_i(s^t)\}_{i=1}^I$ for all s^t , such that:

- (a) The representative consumer and representative firm solves their optimization problem given the price vector.
- (b) All markets clear,

$$\begin{aligned} s_t(z^t) &= 1, \\ b_t(z^t, z_{t+1}) &= 0 \forall z_{t+1} \in Z \\ c_i(z^t) + i_t(z^t) &= y_t(z_t) \\ \forall t, \quad \forall z^{t-1} \in Z^t, \quad \forall z^t \in Z^{t+1}, \quad z_t &\in Z \end{aligned}$$

(d) The first order conditions from the consumer problem is:

$$\begin{aligned} \beta^t \frac{1}{c_t} \pi(s^t) &= \theta_t \\ \theta_t q_t &= \theta_{t+1} [\mathbf{1}(y_t = y_l)] \\ \theta_t p_t &= (y_{t+1} + p_{t+1}) \theta_{t+1} \end{aligned}$$

Substituting for θ_t and θ_{t+1} in the first equation above gives us,

$$c_{t+1} = \beta \pi(s^{t+1} | s^t) c_t \left[\frac{y_{t+1} + p_{t+1}}{p_t} \right]$$

and

$$c_{t+1} = \beta \pi(s^{t+1} | s^t) \frac{c_t}{q_t} \mathbf{1}(y_t = y_l)$$

Note that these are the function of the states, and the price mentioned above are for particular history (s^t). The above equation implies that the consumer equates the marginal benefit from saving to expected discounted marginal return from the savings.

- (e) Using the last two equation from above, we can write the expression for prices for a particular state s^t , that is

$$p_t(s^t) = \pi(s^{t+1}|s^t) \left[\beta \frac{c_t}{c_{t+1}} (y_{t+1} + p_{t+1}) \right]$$

$$q_t(s^t) = \pi(s^{t+1}|s^t) \left[\beta \frac{c_t}{c_{t+1}} (\mathbf{1}(y_t = y_l)) \right]$$

The above are the equilibrium state-contingent asset prices.

- (f) The price $q(s^t)$ is zero for the high state and is equal to $\pi(s^{t+1}|s^t) \left[\beta \frac{c_t}{c_{t+1}} \right]$ for the low state. This is because the this asset pays only if the low state is observed.

Whereas the price of the share is $\pi(s^{t+1}|s^t) \left[\beta \frac{c_t}{c_{t+1}} (y_{t+1} + p_{t+1}) \right]$, this can be seen that the price of the share increases in the probability of high state realization. That is the price of the share will be higher if the expected output at $t + 1$ is higher, thought discounted.

This is intuitive as, in case of the insurance, the price of the insurance should be zero for the case when insurance does not pay in positive. Also, the share price is higher for the case when the expected output is higher, or the probability of high state is higher. If we consider the case, when the y_t follows a Markov process with the absorbent state of high (low), the prices of the share will be increasing (decreasing) with respect to time.

- (g) The unconditional average return of the share is the expected dividend yield. It depends on the expected output growth, and the price of the share, which can be solved solving the equation below.

$$p_t = E_t \left(\beta \frac{c_t}{c_{t+1}} (y_{t+1} + p_{t+1}) \right) \implies 1 = E_t \left(\beta \frac{c_t}{c_{t+1}} \frac{y_{t+1} + p_{t+1}}{p_t} \right)$$

Where, $R_{t+1}^s = \frac{y_{t+1} + p_{t+1}}{p_t}$

Hence, $1 = E_t \left(m_{t+1} R_{t+1}^s \right)$

Therefore, taking expectations both side and using law of iterated expectations, the unconditional return is R_t^s that solves $1 = E(m_t R_t^s)$.

And from the insurance it is,

$$q_t = E_t \left[\beta \frac{c_t}{c_{t+1}} (\mathbf{1}(y_t = y_l)) \right] \implies 1 = E_t \left[m_{t+1} \frac{(\mathbf{1}(y_t = y_l))}{q_t} \right] \implies 1 = \frac{(\mathbf{1}(y_t = y_l))}{q_t} E_t(m_{t+1})$$

Taking expectation both side, and by the law

of iterated expectation, the unconditional return is that solves
 $1 = R_t^i E(m_t)$, Where, $R_t^i = E\left(\frac{\mathbf{1}(y_t=y_l)}{q_t}\right)$

13.1 Solution 1

- (a) To define competitive equilibrium for the Huggett model, we need to add one more financial market (and hence one more asset, linearly independent) to make this a complete market.

A competitive equilibrium in this economy would be a set of value function $V(y, \psi, \eta)$, a decision rule $a' = g(y, \psi, \eta)$, a price system $p = \{R(y', \psi', \eta'), q(y, \psi, \eta; y', \psi')\}$ and the distribution ψ .

Such that:

- Given R and q , consumer optimizes solving the consumer problem:

$$V(y, \psi, \eta) = \max_{a' \geq -\phi} \{u(c) + \beta \sum_{y'} \pi(y'|y) V(y', \psi', \eta')\}$$

s.t.

$$c + \sum_{y' \in Y} q a'(y') + k' = \eta$$

$$\eta' = a'(y', \psi') + (R)k' + y'$$

- Stationarity

$$\psi(B) = \int_S P(x, B) d\psi(x) \quad \forall B \in \mathcal{B}_S$$

- Market clears.

$$\int_S g(y, \psi, \eta) d\psi = 0$$

Walras' law holds and other market clears as well.

Solving the competitive equilibrium above will lead us to the complete market allocation and interest rate. Which we know is $r = 1/\beta - 1$.

There will be no self-insurance, and full risk-sharing. Though the price of the asset will be:

$$q = \pi(y'|y) \beta \frac{u'(c')}{u'(c)} = \pi(y'|y) \beta \frac{c'^\gamma}{c^\gamma}$$

- (b) To solve the incomplete market version of the model as described in the Huggett's paper. This algorithm involves solving through two loops, one on the r and other on the fixed r to solve the consumer Bellman equation, VFI.

Step 1: Discretize the state space (asset space) into $\mathcal{A}$. In discretizing, assign the minimum value as $-\phi$, and the maximum value sufficiently high.

Step 2: For a fixed r , solve the unconstrained consumer problem, through value function iteration over each $a_i \in \mathcal{A}$. If $a' \notin \mathcal{A}$, interpolate the value of $v(a', y')$. Since we exposed no borrowing constraint, if the borrowing condition does not hold, impose $a' = -\phi$.

Step 3: Iterate on r by repeating the step 1 and step 2 until the market clears, i.e., $Ea(r) = 0$. This is the outer loop. For this, we can exploit the Fixed Point problem through which we would be able to find the discrete pdf ψ . This can be achieved by solving the system of linear equation such that $\psi = \psi \mathbf{P}$. If it turns out that $a' \notin \mathcal{A}$, then we can re-weight and split the probabilities through piece-wise linear interpolation. Which will give $a_{i+1} - g(a, y)$ part of $a_{i+1} - a_i$ to the transition probability for (a_i, y') and rest of the weight, i.e., $g(a, y) - a_i$ to the (a_{i+1}, y') .

- (c) I have used the price of the asset rather than return, given the confusion between annualized return. I checked the Huggett's paper, it reports both r and q , my steady state equilibrium q matches with the table 1 of the Huggett's paper.

The figure below shows that the policy function for the low type is concave near the borrowing constraint limit. This implies the lower marginal propensity to save. This can also be seen in the policy function for consumption that the marginal propensity to consume is higher for the low type agents.

- (f) Solving for the steady state price q , I got for the parameters: $(\sigma, \beta, \phi) = (1.5000, 0.9932, -4.0000)$. The equilibrium $(q, Ea(q)) = (0.9980, -0.0070)$. It can be seen that the asset demand is very close to zero.

- (d) As seen in the part (f) above, the asset demand is indeed zero for the 2.3% interest rate or $q = 0.9980$.

- (g) As seen in the graphs, the distribution converges starting from the different asset holdings. This is interesting as it shows that although Huggett's model does not have ex-ante heterogeneity, any ex-ante heterogeneity in wealth will disappear after a few periods.

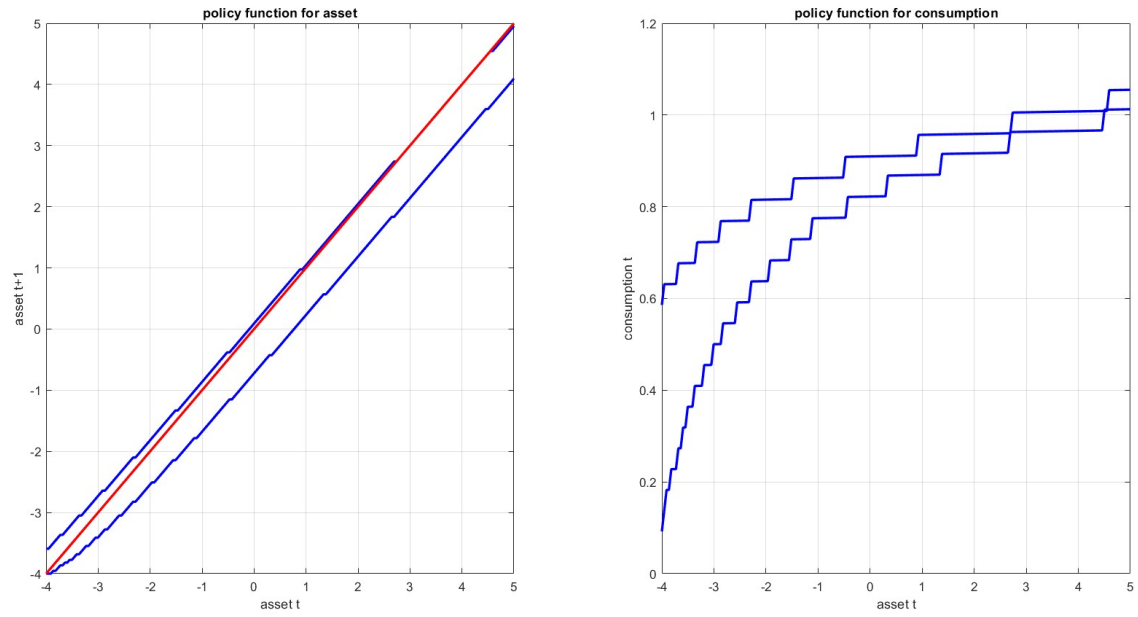


Figure 6:

As we can see both agents that starts soon end up on the same wealth/savings path.

- (h) For the case when $\phi = 6$ and $\sigma = 3$. The figure is shown below. As σ is increased from 1.5 to 3, all individuals become more risk averse. That is their demand for the bond increases, as they want to smooth consumption even more. Therefore, at the previous equilibrium interest rate (1.2%) we would have excess demand. To have equilibrium, the interest rate would need to decrease, implicitly increasing the price of bonds. Interestingly, agents become so risk averse that we need a negative interest rate to incentivize selling bonds. We found an equilibrium interest rate of -4.4%, which is very close to the interest rate in the replication paper cited in part a) and similar to Huggett in that it is negative. In contrast to question h,i) the distribution of assets is not changed much. There is a higher concentration around $a = 1$, but a much lower one in the interval $[2.5, 3.5]$. This is most likely due to the increased risk aversion.

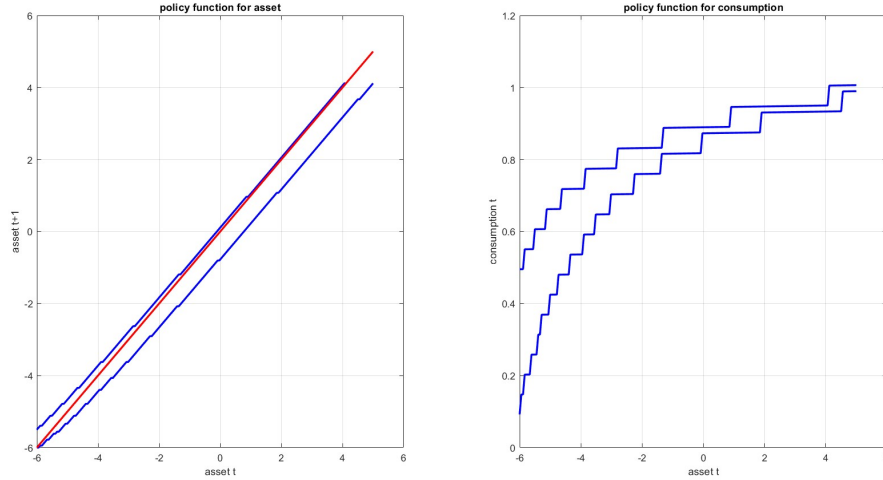


Figure 7:

The distribution of asset is also stationary).

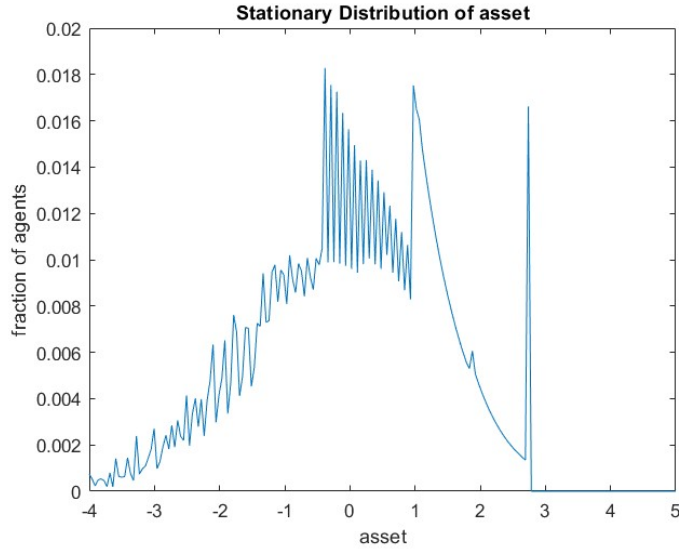


Figure 8:

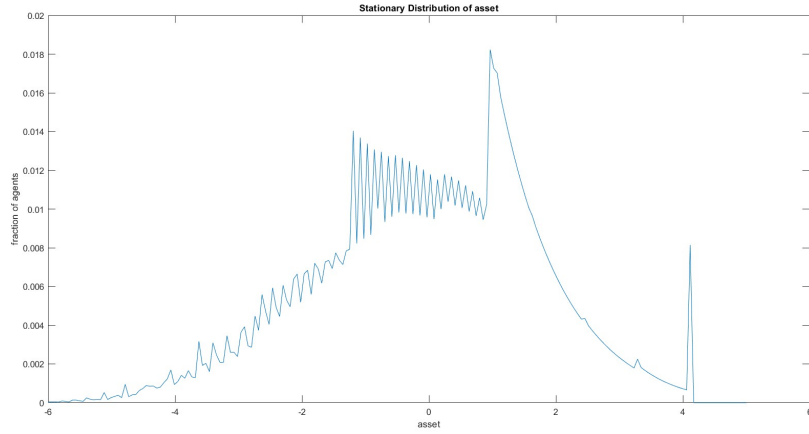


Figure 9:

- (i) This can be seen from the above graphs that the marginal propensity to consume for the case when the agents are more risk averse, the marginal propensity to consume is higher for the individual near the borrowing constraint. Though, the graph flattens out in both of the case, this might be mainly because of the more risk aversion and relaxing the borrowing constraint.
- (j) Solving for the complete market will yield $1 + r = \beta^{-1} = 1.0068$.

To report the difference in the consumption equivalent in the two steady

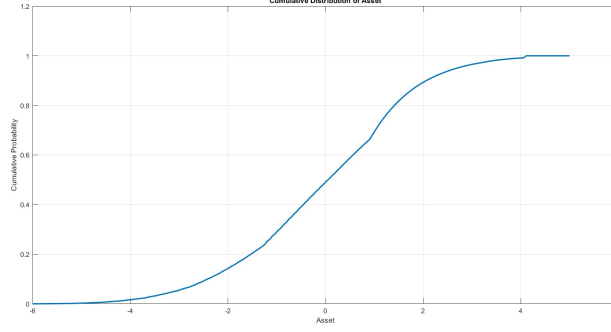


Figure 10:

state, we can use the $1 -$ factor for in the following equality.

$$\int_s E_0 \beta^t u((1 - \lambda) c_t^{inc}) d\psi^{inc} = \int_s v^{comp}(a_0, y_0) d\psi^{comp}$$

Where the superscript "inc" stands for incomplete market, while "comp" stands for complete markets. Following the slides, we know that with a CRRA utility function λ takes the form:

$$\lambda = \left[\frac{\int_s v^{comp}(a_0) d\psi^{comp}}{\int_s v^{inc}(a_0) d\psi^{inc}} \right]^{\frac{1}{1-\sigma}} - 1$$

We can calculate the numerator easily. We have found in part a) that with complete markets: $c = 0.883$ in every period, states and for every individual. Recall, that $\int_s d\psi^{comp} = 1$

$$\int_s v^{comp}(a_0) d\psi^{comp} = v^{comp}(a_0) = \sum_{t=0}^{\infty} \beta^t u(\bar{c}) = \frac{\bar{c}^{1-\sigma} - 1}{\frac{1-\sigma}{1-\beta}} = \frac{-2(0.883^{-0.5} - 1)}{1 - 0.9932} = -18.8797$$

The denominator is calculated using a simulation. To estimate $d\psi^{inc}$, we iterate on the markov chain to find y , then the agent chooses an a' given the current period a , and the policy function. We then evaluate the value function at a , and y . This gives us an empirical density function, where every element of the simulation has a probability of $1/T$.

$$\int_s v^{inc}(a_0) d\psi^{inc} = \sum_{t=1}^T \frac{1}{T} \hat{v}_t^{inc}$$

Where the subscript t , is the element of the simulation. In my calculation i use T of one million and dump the first 10 thousand observations, to remove any effects of a_0 . I find a value of $\sum_{t=1}^T \frac{1}{T} \hat{v}_t^{inc} =$

$$-20.7927.$$

$$\lambda = \left[\frac{-18.8797}{-20.7927} \right]^{\frac{1}{1-1.5}} - 1 = 0.21291843534$$

Solution 2

- (a) For the utility to be an increasing function of the consumption. b need to be very high, i.e., $b < \frac{1}{c}$. Also, b need to be positive as if it is negative then the diminishing marginal utility would not hold.

The Inada condition would never hold, no matter whether the b is positive or negative. As $\lim_{c \rightarrow 0} u'(c) = 1$.

In the case of the positive b and less than the $\frac{1}{c}$, there would be satiation point at $c = 1/b$. Also,

- (b) The relative risk aversion $= -\frac{cu''(c)}{u'(c)} = \frac{cb}{1-cb}$. This can be seen from the expression that increasing consumption would increase the relative risk aversion. Thought, the relative risk aversion will be positive for the case when $c \in (0, 1/b)$.
- (c) Euler Equation: Since the agent faces stochastic income sequence. Suppose the state of the world are captured in the history of events z^t at time t . The consumer solves the following problem:

$$\max_{c_t, a_{t=+1}} E_0 \sum_{t=0}^{\infty} \beta^t u(c_t)$$

s.t.

$$a_{t+1} + c_t = y_t + (1+r)a_t \quad \forall t \quad \forall z^t$$

$$a_0 > 0$$

$$r > 0$$

Setting up the Lagrangian, we have:

$$\mathcal{L} = E_0 \sum_{t=0}^{\infty} \beta^t u(c_t(z^t)) + \sum_{t=0}^{\infty} \lambda_t [y_t(z^t) + (1+r)a_t(z^t) - c_t(z^t) - a_{t+1}(z^t)]$$

First order conditions:

$$\begin{aligned} \frac{\partial \mathcal{L}}{\partial c_t(z^t)} &= \beta^t \pi_t(z^t) u'(c_t(z^t)) - \lambda_t(z^t) = 0 \\ \frac{\partial \mathcal{L}}{\partial a_{t+1}(z^t)} &= \lambda_t(z^t) - \sum_{z^{t+1}} \lambda_{t+1}(z^{t+1})(1+r) = 0 \\ &\forall t \quad \forall z^t \end{aligned}$$

Combining both of the first order conditions, we have

$$\beta^t \pi_t(z^t) u'(c_t(z^t)) = \sum_{z^{t+1}} \beta^{t+1} \pi_{t+1}(z^{t+1}) u'(c_{t+1}(z^{t+1})) (1+r)$$

$$\begin{aligned} \implies u'(c_t) &= \beta(1+r) E_t(u'(c_{t+1})) \\ 1 - bc_t &= \beta(1+r) E_t(1 - bc_{t+1}) \end{aligned}$$

Interpretation: This implies that the consumer equalize the marginal cost of consumption to discounted expected marginal benefit from saving.

Using $\beta^{-1} = 1+r$

$$c_t = E_t(c_{t+1})$$

This is interesting, the above equation is Martingale. The conditional expectation of tomorrow consumption is equal to today's consumption.

- (d) Life time budget constraint: The sum of stream of consumption should be equal to the sum of total income. Writing down the constraint at time period t , we have:-

$$\begin{aligned} c_t + a_{t+1} &= (1+r)a_t + y_t \\ a_t &= \frac{1}{1+r} [c_t - y_t + a_{t+1}] \\ a_{t+1} &= \frac{1}{1+r} [c_{t+1} - y_{t+1} + a_{t+2}] \\ a_{t+2} &= \frac{1}{1+r} [c_{t+2} - y_{t+2} + a_{t+3}] \end{aligned}$$

substituting a_{t+1} in a_t and then a_{t+2} in a_t

$$\begin{aligned} a_t &= \frac{1}{1+r} \left[c_t - y_t + \frac{1}{1+r} [c_{t+1} - y_{t+1} + a_{t+2}] \right] \\ a_t &= \frac{1}{1+r} [c_t - y_t] + \frac{1}{(1+r)^2} [c_{t+1} - y_{t+1}] + \frac{1}{(1+r)^2} a_{t+2} \\ a_t &= \frac{1}{1+r} [c_t - y_t] + \frac{1}{(1+r)^2} [c_{t+1} - y_{t+1}] + \frac{1}{(1+r)^3} [c_{t+2} - y_{t+2}] + \frac{1}{(1+r)^3} a_{t+3} \end{aligned}$$

In the limit-

$$a_t = \left[\sum_{s=0}^{\infty} \frac{c_{t+s}}{(1+r)^{s+1}} - \sum_{s=t}^{\infty} \frac{y_{t+s}}{(1+r)^{s+1}} + \lim_{s \rightarrow \infty} \frac{a_{t+s+1}}{(1+r)^{s+1}} \right]$$

as per the NPG conditions:

$$\begin{aligned} \lim_{s \rightarrow \infty} \frac{a_{t+s+1}}{(1+r)^{s+1}} &= 0 \\ \sum_{s=0}^{\infty} \frac{c_{t+s}}{(1+r)^s} &= \sum_{s=t}^{\infty} \frac{y_{t+s}}{(1+r)^s} + (1+r)a_t \end{aligned}$$

(e) From the Euler equation,

$$\frac{E_t(1 - bc_{t+1})}{1 - bc_t} = \frac{1}{\beta(1+r)}$$

with $\beta(1+r) = 1 \implies c_t = E_t[c_{t+s}] \forall s > 0$. Substituting the value of c_t from the value of part above, we have:

$$\begin{aligned} c_t \sum_{s=0}^{\infty} \left(\frac{1}{1+r} \right)^s &= (1+r)a_t + \sum_{s=0}^{\infty} \left(\frac{1}{1+r} \right)^s E_t(y_{t+s}) \\ c_t \frac{1}{1 - \frac{1}{1+r}} &= (1+r)a_t + \sum_{s=0}^{\infty} \left(\frac{1}{1+r} \right)^s E_t(y_{t+s}) \\ c_t &= r \left[a_t + \frac{1}{1+r} \sum_{s=0}^{\infty} \left(\frac{1}{1+r} \right)^s E_t(y_{t+s}) \right] \\ c_t &= r \left[a_t + w_t \right] \end{aligned}$$

This express the consumption as a fixed marginal propensity to consume r . Here consumption only depends on the expected present value of non-financial income.

(f) The difference between the consumption at $t+1$ to t , with the property of y that $y_t = y + \epsilon_t$

$$\begin{aligned} c_{t+1} - c_t &= r(a_{t+1} + w_{t+1}) - r(a_t + w_t) \\ c_{t+1} - c_t &= r(a_{t+1} - a_t) + r(w_{t+1} - w_t) \end{aligned}$$

Substituting the values of a_{t+1} , w_{t+1} , and a_t

$$\begin{aligned}
\Delta c_t &= r(y_t + (1+r)a_t - c_t - a_t) + r \left[\frac{1}{1+r} \sum_{s=0}^{\infty} \left(\frac{1}{1+r} \right)^s E_{t+1}(y_{t+1+s}) - \frac{1}{1+r} \sum_{s=0}^{\infty} \left(\frac{1}{1+r} \right)^s E_t(y_{t+1+s}) \right] \\
&= r(y_t + ra_t - c_t) + r \left[\frac{y_{t+1}}{1+r} - \frac{y_t}{1+r} \right] \\
&= r(y_t - rw_t) + \frac{r}{1+r} [y_{t+1} - y_t] \\
&= r \left(y_t - \frac{r}{1+r} y_t - r \left(\frac{1}{1+r} \sum_{s=1}^{\infty} \frac{y}{(1+r)^s} \right) \right) + \frac{r}{1+r} [y_{t+1} - y_t] \\
&= \frac{r}{1+r} [y_t - y + y_{t+1} - y_t] \\
&= \frac{r}{1+r} [y_{t+1} - y]
\end{aligned}$$

This can be seen from the last expression that the consumer would like to smooth the consumption, and therefore take into account the deviation of the income in the next period from the mean. This can also be interpreted as, the difference between tomorrow's consumption to today is equal to βr times the deviation of income from mean tomorrow.

- (g) The change in the credit constraint will make the agent to not borrow till a certain limit, which is $\underline{a}$ here. Therefore, for the agents for whom the borrowing constraint will be binding, they will

Solution 3

- (a) The natural borrowing limit for bonds is something the minimum the individual will earn over the time period.

$$\begin{aligned}
a_{t+1} &\geq \sum_{i=t}^{\infty} \frac{c_i - ws}{(1+r)^i} \\
&\geq \sum_{i=t}^{\infty} \frac{ws}{(1+r)^i} \\
&\geq - \sum_{i=t}^{\infty} \frac{ws_{min}}{(1+r)^i} \\
&= - \frac{ws_{min}}{r}
\end{aligned}$$

Therefore the natural borrowing limit is

$$\phi = \min\{\underline{b}, \frac{ws_{min}}{r}\}$$

- (b) The consumer will choose the quantity of bond and capital that maximizes the life-time utility. The consumer problem is given follows:

$$v(\omega, s) = \max_{b', k'} \left\{ u(c) + \beta \sum_{s' \in S} \pi(s'|s) v(\omega', s') \right\}$$

subject to:

$$\begin{aligned} c + qb' + k' &= \omega \\ \omega' &= ws' + (1 + r^k)k' + b' \\ b' + k' &\geq -\phi \quad (\text{Borrowing constraint}) \end{aligned}$$

Solving the problem will give the portfolio choice, i.e., (b, k) .

- (c) Solving for the first order conditions for the consumer problem will give us two sets of FOCs, that are:

$$u'(c) = \beta(1 + r^k) E_t u'(c')$$

and

$$u'(c) = \frac{\beta}{q} E_t u'(c')$$

For these two Euler equations to hold, we need to have in equilibrium $(1 + r^k = 1/q)$. Therefore the consumer problem can now be written as:

$$v(\omega, s) = \max_{a'} \left\{ u(c) + \beta \sum_{s' \in S} \pi(s'|s) v(\omega', s') \right\}$$

subject to:

$$\begin{aligned} c + qb' + k' &= ws + (1 + r^k)k + b \\ c + \hat{b} + k' &= ws + (1 + r^k)k + \hat{b}/q \\ c + a' &= ws + (1 + r)a \quad \text{keeping } a = k + b \\ b' + k' &\geq -\phi \quad (\text{Borrowing constraint}) \end{aligned}$$

By this the individual-level portfolio composition is not defined all the time, though in the aggregate we know what $a' = b' + k'$

is. In the case of negative a' , we get to know that the capital can not be negative, therefore $b' < -k'$.

- (d) The recursive competitive equilibrium (RCE) is a value function $v(\omega, s)$, the decision rule $a' = g(\omega, s)$, prices w and r , and the distribution ψ , such that:

- 1) Given r and w , the consumer solves the problem (defined above).
- 2) Given r and w , the representative firm solves the problem such that:

$$r = F_K(K, N) - \delta$$

$$w = F_N(K, N)$$

- 3) All markets clear:
 - $\int_S g(\omega, s) = K + b = K$ capital market, as $b = 0$ in equilibrium.
 - $N = \sum_s \Pi(s)$ Labor market clears (LLN)
- 4) The distribution over the state space is stationary, i.e.,

$$\psi(B) = \int_S P(x, B) d\psi(x) \quad \forall B \in \mathcal{B}_S$$

Where, P is transition matrix generated from the $\pi(s'|s)$

- (e) Tightening of the credit constraint will increase the aggregate capital (K). Why? This is because of the precautionary savings. The agents that are near to the credit limit, will have very less marginal propensity to save. The change in the borrowing constraint is a kind of MIT shock. This will induce the transitional dynamics and then the economy will converge to steady state. This transition will include agent near to the limit $-\phi$ saving less, as their marginal propensity to save is very low. For the same interest rate, the individual will supply more capital and demand bond. Inter-temporal effect and precautionary savings comes into play. Due the high supply of the capital for a fixed price r , this will shift the aggregate capital supply curve to the right, which will induce the lower equilibrium price r . This leads to the increase in the $K_{IM} - K_{CM}$. This implies, the idiosyncratic risk matters for the macro aggregates, and there will be over accumulation of the output, and hence inefficient.

- (f) The change of aggregate capital (positive) due to tightening the credit constraint may not be welfare improvements. Output is not welfare. There will be a lot of distributional changes, the distribution ψ will change to ψ' . Comparing the steady state welfare change will be easier to compute, but is very hard to interpret the welfare effects. This is because, comparing steady state ignores the transition dynamics.